



Autonomous Unmanned Aircraft Systems for Enhanced Search and Rescue of Drowning Swimmers: Location-Allocation Optimization and Flight Trajectory Planning

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Abstract

Drowning is among the most prevalent causes of death from unintentional injuries worldwide. Because of the time-sensitive nature of swimming accidents and the shortage of lifeguard staff resulting in unsupervised swimming areas, interest in innovative supportive rescue methods increases. In this paper, we introduce an autonomous Unmanned Aircraft System (UAS) that can be employed by Emergency Medical Service (EMS) providers in swimmer rescue scenarios additionally to Standard Rescue Operation (SRO) equipment. The UAS consists of Unmanned Aerial Vehicles (UAVs) and purpose-built hangars located near the swimming area to store the UAVs. When receiving an alert, the UAVs autonomously navigate to the emergency site to conduct a Search and Rescue (S&R) operation for the swimmer in distress. We introduce a Mixed-Integer Linear Programming (MILP) model to address the Facility Location-Allocation Problem (FALP), identifying optimal hangar locations and allocation of UAVs to hangars to serve the demand in the swimming area. Additionally, we present a MILP model to optimize the UAV flight trajectories in advance of the operation, allowing for efficient coordination of a heterogeneous UAV swarm. We apply the presented MILP models to a real-world scenario in the Lusatian Lake District, Eastern Germany using the state-of-the-art commercial solver CPLEX to solve the instances.

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1 Introduction

Swimming is a popular sport and also a recreational activity known for its benefits in improving physical and mental health and is frequently recommended to medical patients (Petrescu et al. 2014). One significant advantage of swimming is the ability to engage this sport in various settings such as pools, lakes, rivers and seas with minimal equipment required (Barghamadi and Anvari 2019). However, swimmer safety is substantial, as one omnipresent danger is drowning, the process of experiencing respiratory impairment from immersion or submersion in liquid (Idris et al. 2014). Drowning ranks as the third leading cause of unintentional injury deaths worldwide, with the World Health Organization (WHO) estimating approximately 236,000 annual deaths and due to unreported accidents, this number may be even significantly higher (World Health Organization 2023). The largest voluntary water rescue organization in the world, German Life Saving Association (DLRG - Deutsche Lebens-Rettungs-Gesellschaft), recorded 378 fatal drowning accidents in Germany in 2023 showing a rising trend compared to previous years (2022: 355, 2021: 299). Notably, 90 % of these incidents occurred in inland waters like rivers, lakes, streams, ponds and canals. Coastal areas are usually well supervised, but other swimming areas often remain inadequately or even completely unguarded due to staff shortage (DLRG 2023).

Drowning is typically a silent event, where victims rarely exhibit convulsive or panicked movements. Instead, they conserve their energy, attempting to keep their heads just above the water's surface for as long as possible. This often results in victims being unable to draw attention to themselves or call for help (Jalalifar et al. 2024). Consequently, drowning incidents often go unnoticed in large inland waters without professional lifeguard supervision. Even when noticed, standard rescue operations (SRO) utilizing conventional waterway equipment such as lifeboats and lifebuoys may be ineffective due to the time-sensitive nature of drowning accidents and the significant delay until the equipment is prepared and the external rescue staff arrives on site. Technical innovations, such as Unmanned Aerial Vehicles (UAVs), often colloquially referred to as drones, have the potential to provide a remedy.

Our idea envisions an Unmanned Aircraft System (UAS) or drone-in-a-box system that consists of a UAV swarm allocated to purpose-built UAV hangars to provide rapid first response in a drowning emergency, applying it specifically to a swimming area within an inland water body. UAVs offer advantageous bird's-eye views and a broad variety of integrated flight sensors such as Global Positioning System (GPS), acceleration and yaw rate sensors, barometers, magnetometers, cameras with polarization filters, sonar, LiDAR, infrared (IR) or radar. Their fast deployment and high degree of automation have great potential to improve the quality of medical care. We station the hangars, which are intended to maintain the UAVs highly automated, on land near the swimming area and equip them with one or even multiple UAVs. In case of

emergency, after being alerted by bystanders or the distressed swimmer himself, the Emergency Medical Service (EMS) provider can dispatch the UAS in addition to SRO resources. The UAVs will then autonomously perform a Search and Rescue (S&R) operation in a predefined target area. The UAVs are equipped with a camera sensor and flotation devices and use computer vision algorithms to automatically locate and identify distressed swimmers and drop off the flotation device. This aims to reduce the treatment-free interval and mitigate long EMS response times for large, partially inaccessible and unsupervised swimming areas with low personnel expenses.

This paper is part 1 of a two-part series of papers. In this paper, we focus on planning flight trajectories of UAV swarms for S&R operations on inland water surfaces to locate the target as quickly as possible. For this, we introduce a Mixed-Integer Linear Programming (MILP) model. This approach allows the incorporation of preliminary information about the possible location of the swimmer to be found as well as compliance with restricted areas and the influence of wind. Additionally, we introduce a MILP model to simultaneously find the optimal locations of a selection of UAV hangars, as well as the configuration of the stored UAV swarm. This model is designed to ensure time-wise and battery-wise coverage of the operational area. In part 2, we examine the discrete-event simulation (DES) of UAS-assisted water rescue missions in order to compare the reaction times for different UAS configurations using the calculation results of the location model and the flight model from part 1. We further introduce a You Only Look Once (YOLO) (Redmon et al. 2016; Terven et al. 2023; Mohsen Khan Mohammadi et al. 2024) neural network framework to detect distressed swimmers by camera from UAV perspective in real-time.

This research was carried out as part of the *RescueFly* research project, which aimed to develop a prototype UAS for a test area in the Lusatian Lake District that lies partly in Brandenburg and partly in the Free State of Saxony in Germany. Figure 1a depicts a purpose-built water rescue UAV, manufactured by MINTMASTERS GmbH, equipped with self-inflatable flotation device *Restube* (Restube GmbH 2024), and Figure 1b shows UAV hangar *dRack*, manufactured by DELTA-Fluid Industrietechnik GmbH, with integrated computer technology, software server and sensor technology from Chemnitz University of Technology (Heller et al. 2024). Figure 1c shows the advantageous bird's-eye view of a UAV for search missions at a popular local beach in the *RescueFly* test area in Partwitz, Germany. This paper does not delve into regulatory constraints that may affect UAV deployment in actual emergency scenarios, technical details about UAV or UAV hangar manufacturing, flight safety or cost-effectiveness and resource requirements of implementing a UAV-based water rescue system like UAV maintenance, personnel training or infrastructure setup.

The remainder of the paper is organized as follows. Section 2 reviews the most relevant literature to our contribution, mainly focusing on UAV flight path planning for maritime S&R operations and UAS location-allocation optimization. Section 3 introduces a MILP approach for UAV flight trajectory optimization for swimmer S&R. In Section 4, a MILP approach for UAV hangar and UAV swarm location-allocation optimization is presented. Section 5 presents a case study of an operational area consisting of multiple large lakes in the Lusatian Lake District, Germany. Finally, Section 6 contains a conclusion and a discussion of future research topics.



(a) Water rescue UAV prototype "BUDDY I" equipped with a Restube. Photograph by Björn-Steiger-Stiftung, (2023).



(b) RescueFly's UAV hangar prototype. Photograph by Ralf Schuster, BTU Cottbus-Senftenberg, (2023).



(c) Bird's eye view of Lake Partwitz, Lusatian Lake District. Photograph by Björn-Steiger-Stiftung, (2023).

Fig. 1 The RescueFly UAS setup including the UAV, hangar and the operational environment

2 Literature Review

Recently, there have been numerous promising inventions to enhance swimmer safety, that can be categorized as follows (Samuel Ndueso John et al. 1987):

1. *Physical barriers*: These are typically fences or gates for home pools to enhance safety especially for children (Bugeja et al. 2013).
2. *Detection devices*: Mainly encompassing computer vision and deep learning approaches, these devices are used to detect drowning individuals. Examples are surveillance cameras at sea (Li et al. 2021) and over- and underwater cameras in pools (He et al. 2023; Hasan et al. 2021; Lei et al. 2022).
3. *Signaling devices*: Wearable health monitoring systems (WHMS) such as wristbands (Samuel Ndueso John et al. 1987) or headbands (Molly et al. 2022), which utilize biosensors to assess health parameters such as heart rate, blood pressure, respiration rate and skin temperature (Pantelopoulous et al. 2010).
4. *Flotation devices*: Inflatable wristbands, life-ring Unmanned Aerial Vehicles (UAVs), Unmanned Sea Vehicle (USV) delivery systems or garment-integrated personal flotation devices (Barghamadi and Anvari 2019; Lygouras et al. 2017; Geert De Cubber et al. 2013).
5. *Devices for emergencies involving water sport events*: These are mainly protective clothing for water sports, such as protective helmets for water skiing or wakeboarding (Scher et al. 2020).
6. *Devices for preventing drowning in home sanitary wares*: These typically use robotic technology such as hybrid camera and overhead gantry robots (Jose and Udupa 2021) or robotic underwater lifts (Laxman and Jain 2022).
7. *(Scuba) Diving devices*: Diving inventions such as survival suits that can turn divers to a face-up floating position, inflatable bladders from hand-held triggers, automatic alarm systems to release sparklers and dyes (Gunatilaka and Ozanne-Smith 2004).
8. *Other miscellaneous inventions*: Unlocking devices for motorboats after immersion in water, low friction liners in waders, etc. Gunatilaka and Ozanne-Smith (2004).

Our proposed UAS mainly falls into categories (2) and (4) as we aim to detect distressed swimmers and drop flotation devices by UAV but there are also possible

synergies with (3) and (5) when alerting is automated via WHMS and the target group is expanded from swimming to water sport in general. For further readings on recent drowning prevention technologies we refer to the review articles of Jalalifar et al. (2024); Angela and Beale-Tawfeeq (2019) and Pantelopoulos et al. (2010). Furthermore, Yeong et al. (2015) substantiate UAV applicability to maritime S&R operations and Otto et al. (2018) provide an entry point into the topic of civil applications of UAVs, focusing on planning search operations and routing for a set of locations. In the following, we review the most important and popular works that are relevant to our contribution, mainly focusing on UAV flight planning and UAS location-allocation optimization.

2.1 UAV Flight Path Planning

The potential of UAVs supporting S&R operations receives growing recognition, as indicated by the increasing number of publications in this field (Waharte and Trigoni 2010). Especially so-called Micro Aerial Vehicles (MAVs) are of great interest due to their good maneuverability, small size and ability to hover, making them valuable for use in areas that are difficult to access (Schmidt and Fügenschuh 2021). Equipped with sophisticated sensors, they are predestined to detect targets, such as swimmers, efficiently. To exploit the full potential of UAVs, the employed search algorithm is crucial to locate the target as quickly as possible. For the design of search algorithms, the following fundamental parameters need to be considered: The quality of sensory data collected by UAVs, battery limitations, environmental factors such as wind and obstacles and the level of information exchange between UAVs (Waharte and Trigoni 2010). Several S&R methods have been developed to date. The common framework for an S&R algorithm is to divide the search area into grid cells and assign probability values to each cell indicating the likelihood that the target is located there. Utilizing this framework, Waharte and Trigoni (2010) compared real-time greedy, potential-based and partially observable Markov Decision Process-based heuristics, considering UAV information sharing and stationary targets through simulation. They testified that real-time approaches might be more appropriate because they can cope with uncertainties of real-world deployment. Pitre et al. (2012) developed a method for joint UAV S&R operations route planning, also considering a cell- and probability map-based approach and particle swarm optimization for large-scale search areas and moving targets. Hayat et al. (2017) introduced a hybrid Multi-Objective Path Planning (MOPP) genetic algorithm (GA) for a UAV swarm to minimize the time to find a target (search task) by either covering an operational area or to ensure connectivity to quickly communicate this information to a ground base station (inform task) by using a UAV mule, relay chain or a hybrid approach. This work was extended by Hayat et al. (2020) and Yanmaz (2022), proposing jointly-optimized multi-UAV path planners called Simultaneous Inform and Connect with quality-of-service (SICq) and dynamic relay positioning algorithm (RPA), respectively (Yanmaz 2023). SICq takes quality-of-service (QoS) into account in form of a monitor task, designing two path planning strategies to firstly optimize search, inform and monitor tasks jointly and secondly optimize search and inform tasks jointly, followed by optimization of monitor tasks.

They illustrate that this joint optimization allows monitor tasks to be performed additionally to search and inform tasks without degrading the overall mission quality. RPA divides the UAV swarm into search UAVs and relay UAVs, optimizing the search paths through minimization of the total mission time. Yanmaz (2023) conducted a comparative analysis of the above mentioned schemes and propose a hybrid approach using SICq for pre-planning for the search UAVs and RPA for the relay UAVs. However, the setting is still far away from reality: A rectangular application area is considered, which is divided into cells. The UAVs can then move from cell to cell at a fixed speed, where they stay for a fixed time to observe the cell in order to find the target object. Important mission- and UAV-specific parameters such as acceleration, battery capacity or restriction areas are disregarded. Kumar et al. (2022) integrated collision and obstacle avoidance in a particle swarm optimization framework for S&R path planning of UAV swarms by changing flight altitudes of respective UAVs accordingly. Ruetten et al. (2020) optimized a UAV swarm to maximize the area being searched while maintaining wireless intercommunication of the swarm within a S&R operation utilizing genetic algorithms (GA). Cho et al. (2021) introduced a randomized heuristic search approach to solve the Coverage Path Planning (CPP) problem which can be regarded as a variant of the classical Vehicle Routing Problem (VRP). They decompose a polygonal search area into an equidistant grid to transform it to a graph of nodes and vertices and develop a Mixed-Integer Programming (MIP) model with the objective of minimizing the mission completion time to cover the whole search area by a swarm of UAVs. They apply that model to a marine field located in the West Coast of Korea and introduce a heuristic that drastically shortens the computing time compared to a commercial MIP solver while still delivering acceptable results. Yuwen (2023) performed UAV swarm S&R path planning by discretizing the search area as grid and subdivision into subareas for each UAV which is assigned a task. Then, an enhanced A^* algorithm is used to find the shortest collision-free paths to a respective target node. Zhan et al. (2024) developed a reinforcement learning-based genetic algorithm (GA-RL) to solve the maritime S&R path planning problem for multiple rescue centers deploying UAVs. They demonstrate the superiority of GA-RL over other outstanding heuristic algorithms and conduct a case study on the South China Sea.

For comprehensive overviews on the broad spectrum of UAV search algorithms and applications to S&R operations, we refer to the review papers of Lyu et al. (2023); Puente-Castro et al. (2022); Ajith and Jolly (2021); Debnath et al. (2019) and Karamanou and Drellosi (2018).

The vast majority of all algorithms for search and rescue optimization use grid or graph structures as a framework, where the searchers can move from cell to cell and observe them for a certain period of time. However, this involves making highly simplifying assumptions, particularly with regard to the flight dynamics of the UAVs. To account for this, we apply the MIP model of Schmidt and Fügenschuh (2023) for general mission and flight planning of UAV swarms to the use case of search and rescue of swimmers. Additionally, this gives us the opportunity to integrate restriction areas, wind and weighted demand locations.

2.2 UAS Location-Allocation Optimization

A suitable location for supply units is crucial, especially in time-critical environments such as EMS. Consequently, considerable efforts have been made to solve various location optimization problems, known in the literature as Facility Location Problems (FLPs), Location-Allocation Problems (LAPs) or Hub Location Problems (HLPs). Laporte et al. (2019) gave a detailed introduction to the field of *location science*, presenting various location problem formulations for diverse applications. Horst et al. (2023) focused on approaches that incorporate uncertainty into the EMS facility location framework. Several review papers concentrate on location problems in the EMS context (ReVelle et al. 1977; Brotcorne et al. 2003; Jeffrey and Goldberg 2004; Li et al. 2011), or encompassing the entire range of EMS systems, considering planning problems throughout the entire emergency care pathway (Aringhieri et al. 2017).

To date, the number of publications on location optimization of UAVs for our specific application, the inland-water S&R, or more generally the maritime S&R, is rather sparse. However, some maritime S&R location and resource allocation optimization publications with conventional rescue means such as vessels and helicopters exist. Sun et al. (2022) introduced a non-linear optimization model to solve the problem of maritime S&R location-allocation in the coastal area of Hainan, China. They consider ships and aircraft and apply GA, PSO and enhanced particle swarm optimization (EPSO) and found that EPSO outperformed its competitors. Pelot et al. (2015) applied classical emergency vehicle location optimization models (MCLP (Church and ReVelle 1974), MEXCLP (Daskin 1983) and CMCLP (Pirkul et al. 1991)) to maritime S&R. They further consider secondary service and uncovered service demand. The models were applied to the maritime area of Eastern Canada, where spatial demand was determined on the basis of historical S&R incidents data. Olgac and Toz (2022) applied the Set Covering Problem (SCP) and MCLP models to maritime UAV-based S&R in the immigration hot spot area Aegean Sea Turkish S&R zone. They further introduce geographical suitability values (GSVs) and meteorological suitability values (MSVs) that are factors affecting the optimum location of the UAV's ground control station (GCS). Akbari et al. (2018) extended the standard p-Median model and MCLP to incorporate multiple facility types and applied it to maritime S&R for Atlantic Canada. Coverage is considered regarding the range and travel time of the deployed vessels. They evaluate and compare the performance of model solutions via simulation, introducing multiple assessment criteria, namely the access time (mean time to reach a demand point), primary coverage (percentage of covered demand points within access time), backup coverage (percentage of incidents within predetermined coverage region of at least two vehicles), the Gini index (deviation level of the access time to all incidents) and the maximum access time (maximum time to reach a demand point). Karatas (2021) introduced an advanced MILP for maritime S&R, considering multiple helicopter and vessel types, demand seasonality and uncertainty, demand types that require specific servers, budgetary limitations, and relocatable and non-relocatable resources. They applied the model to the Aegean Sea area and measured the performance of their solutions with ten different metrics. Ferrari and Chen (2020) introduced a binary integer programming (BIP) model to solve the aerial S&R swarm

resource allocation problem on a tactical planning level, optimizing for the search area to be covered, the time flown and costs for the operations. Braßel et al. (2023) developed a MILP for inland-water UAS location optimization considering multiple wind scenarios to find robust UAS hangar locations as well as possible restricted areas to be avoided by employing a graph-based routing approach. As in other standard models, they consider an individual one-to-one assignment of exactly one server to one demand point. They further only account for one hangar type and one assigned UAV type that serves one single purpose or contribution.

In our model, however, we want to extend this approach by considering multiple different UAV types and hangar types as well as different types of demand of these UAV types are considered. Thus, different types of UAVs can contribute to different mission tasks, such as searching, surveillance or medical supplies transportation. Further, we incorporate a mechanism to enable cooperative missions, as these offer considerable potential for improving the success rate of a S&R mission. Furthermore, hangar- and UAS-specific parameters are integrated, such as a hangar's capacity and start delay and a UAV's size, speed, acceleration and climb rate.

3 Flight Trajectory Optimization of a maritime UAV-based Search and Rescue Operation

S&R has many important use cases including the search for missing people in difficult terrain and weather conditions, in disaster response situations, aerial surveillance or S&R in maritime environments. Each use case has its own characteristics, requiring a compatible search algorithm. Since we focus on the latter use case and restrict our search area to water bodies, we need to choose our search method accordingly. We consider a stationary UAS that is deployed on a ground location. Since our operational area is fixed by the stationary UAS, we opt to use preliminary information about the area, also incorporating UAV flight dynamics to plan the flight precisely in advance. Here, we introduce an offline approach exclusively for mission pre-planning. For practical application, a hybrid architecture that can switch between global deliberative planning in advance and reactive real-time control is necessary to deal with uncertain environments (Yasin et al. 2020).

In the following, we adapt a planar (two-dimensional) version of the UAV flight path planning models from Schmidt and Fügenschuh (2023), Schmidt and Fügenschuh (2021) and Fügenschuh and Müllenstedt (2015). They extended classical Operations Research (OR) problems, namely the *Vehicle Routing Problem (VRP)* and the *Orienteering Problem (OP)*, by considering UAV flight dynamics, battery limitations, wind and restricted areas. In contrast to vehicles moving on a street network modeled by a graph, the UAV can operate freely in the three-dimensional space (Fügenschuh and Müllenstedt 2015). We maintain a fixed flight altitude throughout the entire S&R mission because of legal regulations, flight safety and to guarantee a high detection rate by choosing an optimal flight altitude for a deep learning approach for automatic swimmer localization. Thus, we neglect the third dimension in our approach.

Since this is a variant of the NP-hard OP, it is quite challenging to solve larger instances to proven global optimality (Schmidt and Fügenschuh 2023). An overview

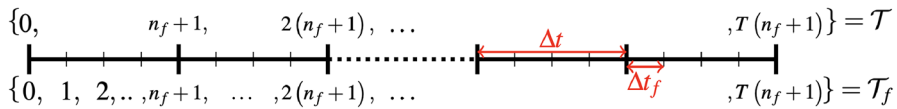


Fig. 2 Visualization of the two time discretization schemes (Schmidt and Fügenschuh 2023)

about OP and VRP and its variants is provided by Vansteenwegen et al. (2011) and Braekers et al. (2016), respectively.

3.1 Model Framework

In the following, the adapted UAV flight planning model is presented, including collision avoidance, restricted areas, wind zones, Newtonian flight dynamics and prioritization of waypoints. Vectors, multi-dimensional decision variables and matrices are written in bold, sets are shown as large italic letters and parameters and one-dimensional decision variables in normal font. We utilize a time discretization, including T time steps and a time step size Δt . We now introduce two different time discretization schemes: a fine one $\mathcal{T}_f = \{0, 1, 2, \dots, Tn_f\}$ and a coarse one $\mathcal{T} = \{0, n_f, 2n_f, \dots, Tn_f\}$ where $n_f \in \mathbb{R}^2$ denotes the number of fine time steps to get a coarse time step and the fine step size Δt_f is defined by $\Delta t_f := \frac{\Delta t}{n_f}$. Note that $\mathcal{T} \subseteq \mathcal{T}_f$ and $|\mathcal{T}| = \text{const.}$ for varying values of n_f . This coupled two-time-level discretization approach introduced by Schmidt and Fügenschuh (2023) helps reducing the number of variables and constraints while still achieving a smooth flight trajectory by controlling n_f .

From a modeling perspective, the two-level discretization scheme separates decision resolution from trajectory resolution. The main binary and coupling decision are only taken at coarse time steps, which keeps the MILP size manageable, while the fine grid $\mathcal{T}_f$ is used to propagate the vehicle dynamics and ensure smooth motion between consecutive coarse decisions. A purely fine discretization would significantly increase the number of variables and constraints, quickly making the model intractable, whereas a purely coarse discretization would lead to non-smooth and physically unrealistic trajectories and poor approximations of the continuous flight dynamics. Continuous-time formulations with checkpoints would require nonlinear or Mixed-Integer Nonlinear Programming (MINLP) models and are therefore considerably harder to solve. The chosen scheme provides a practical compromise that preserves linearity while enabling smooth and physically plausible trajectories at moderate computational cost.

We now introduce a function $\hat{t} : \mathcal{T}_f \rightarrow \mathcal{T}$ of the form

$$\hat{t} = \left\lfloor \frac{t}{n_f} \right\rfloor n_f \quad (1)$$

to map fine time steps to their coarse counterpart by rounding down to the previous coarse time step. Figure 2 visualizes both time discretization schemes $\mathcal{T}$ and $\mathcal{T}_f$.

We denote the set of UAVs as $\mathcal{U}$ and the set of waypoints to visit as $\mathcal{W}$. The set $\mathcal{W}_u \subseteq \mathcal{W}$ determines which waypoints are suitable for the UAV $u \in \mathcal{U}$, depending on

factors such as equipment or allocation. We define the position vector of the respective UAV $u \in \mathcal{U}$ at timestep $t \in \mathcal{T}_f$ as $\mathbf{r}_{u,t} = (r_{u,t}^x, r_{u,t}^y)^T$, where the first component refers to the x -direction and the second to the y -direction. In the following, we will use this convention for each introduced spatial variable. Every UAV $u \in \mathcal{U}$ has a fixed start and end location

$$\mathbf{r}_{u,0} = \mathbf{r}_u^{\text{start}} \in \mathbb{R}^2, \mathbf{r}_{u,n_f T} = \mathbf{r}_u^{\text{end}} \in \mathbb{R}^2 \quad (2)$$

and a ground control station $\mathbf{G}_u \in \mathbb{R}^2$. This is the center of the operation range $\rho_u \in \mathbb{R}_{>0}$ which cannot be exceeded by the respective UAV, ensured by constraints on the position $\mathbf{r}_{u,t}$ of UAV $u \in \mathcal{U}$:

$$\|\mathbf{r}_{u,t} - \mathbf{G}_u\|_2 \leq \rho_u \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f. \quad (3)$$

We introduce binary variables $b_{u,t}^+, b_{u,t}^-, b_{u,t} \in \{0, 1\}$ to track whether a UAV is airborne:

$$b_{u,t}^+ = \begin{cases} 1, & \text{if UAV } u \in \mathcal{U} \text{ takes off before time step } t \in \mathcal{T} \\ 0, & \text{else,} \end{cases} \quad (4)$$

$$b_{u,t}^- = \begin{cases} 1, & \text{if UAV } u \in \mathcal{U} \text{ landed before or at time step } t \in \mathcal{T} \\ 0, & \text{else,} \end{cases} \quad (5)$$

$$b_{u,t} = \begin{cases} 1, & \text{if UAV } u \in \mathcal{U} \text{ is midair at time step } t \in \mathcal{T} \\ 0, & \text{else.} \end{cases} \quad (6)$$

We do not allow intermediate landings, thus it holds

$$b_{u,t+n_f}^- \leq b_{u,t}^- \quad \forall u \in \mathcal{U}, t \in \mathcal{T} \setminus \{n_f T\}, \quad (7)$$

$$b_{u,t}^+ \leq b_{u,t+n_f}^+ \quad \forall u \in \mathcal{U}, t \in \mathcal{T} \setminus \{n_f T\}, \quad (8)$$

$$b_{u,t} = b_{u,t}^+ + b_{u,t}^- - 1 \quad \forall u \in \mathcal{U}, t \in \mathcal{T}. \quad (9)$$

To ensure collision avoidance of the UAVs, we introduce a safety distance $\epsilon \in \mathbb{R}_+^2$ around every UAV and define collision avoidance binary variables $\underline{e}_{u,u',t}, \bar{e}_{u,u',t} \in \{0, 1\}^2$ for $u, u' \in \mathcal{U}, u < u'$ as a vector in x - and y -coordinate as follows:

$$\underline{e}_{u,u',t} = \begin{cases} 1, & \text{if positive distance between } u \text{ and } u' \text{ is smaller than} \\ & \text{safety distance } \epsilon \text{ at time step } t \in \mathcal{T} \\ 0, & \text{else.} \end{cases} \quad (10)$$

$$\bar{e}_{u,u',t} = \begin{cases} 1, & \text{if negative distance between } u \text{ and } u' \text{ is smaller than} \\ & \text{safety distance } \epsilon \text{ at time step } t \in \mathcal{T} \\ 0, & \text{else.} \end{cases} \quad (11)$$

Using the ones vector $\mathbf{1} = (1, 1)^T$, UAV position $\mathbf{r}_{u',t}$, the collision avoidance binary variables $\underline{\mathbf{e}}_{u,u',t}$, $\bar{\mathbf{e}}_{u,u',t}$, and a sufficiently large constant M , the following constraints ensure collision avoidance:

$$\mathbf{r}_{u',t} + \boldsymbol{\varepsilon} - M\bar{\mathbf{e}}_{u,u',t} \leq \mathbf{r}_{u,t} \quad \forall u, u' \in \mathcal{U} : u < u', t \in \mathcal{T}_f, \quad (12)$$

$$\mathbf{r}_{u,t} \leq \mathbf{r}_{u',t} - \boldsymbol{\varepsilon} + M\underline{\mathbf{e}}_{u,u',t} \quad \forall u, u' \in \mathcal{U} : u < u', t \in \mathcal{T}_f, \quad (13)$$

$$\mathbf{1} \cdot \underline{\mathbf{e}}_{u,u',t} + \mathbf{1} \cdot \bar{\mathbf{e}}_{u,u',t} \leq 5 - b_{u,t} - b_{u',t} \quad \forall u, u' \in \mathcal{U} : u < u', t \in \mathcal{T}. \quad (14)$$

Next, we incorporate restricted areas, often also referred to as no-fly zones or geofences, which cannot be entered by the UAVs. By utilizing a set of polyhedral elements $\mathcal{Q}$, it is possible to model complex static or moving geometries by decomposition into multiple convex subparts. Each $q \in \mathcal{Q}$ is described by N_q hyperplanes and is of the form $\{\mathbf{x} \in \mathbb{R}^2 | \mathbf{A}_{q,t}\mathbf{x} \leq \mathbf{c}_{q,t}\}$ with $\mathbf{A}_{q,t} \in \mathbb{R}^{N_q \times 2}$ and $\mathbf{c}_{q,t} \in \mathbb{R}^{N_q}$. We define binary variables $f_{u,q,t} \in \{0, 1\}^{N_q}$ that indicate if a UAV is outside a restricted area with respect to the corresponding hyperplane

$$f_{u,q,t}^k = \begin{cases} 1, & \text{if UAV } u \in \mathcal{U} \text{ is outside restricted area } q \in \mathcal{Q} \\ & \text{w.r.t. hyperplane } k \text{ of } \{1, \dots, N_q\} \text{ at time step } t \in \mathcal{T}, \\ 0, & \text{else.} \end{cases} \quad (15)$$

Utilizing a sufficiently large constant M and binary variable vectors $\mathbf{f}_{u,q,t} = (f_{u,q,t}^1, \dots, f_{u,q,t}^{N_q})^T$, the following constraints on a UAV's position $\mathbf{r}_{u,t}$ ensure staying outside the restricted areas:

$$\mathbf{A}_{q,t}\mathbf{r}_{u,t} \geq \mathbf{c}_{q,t} - M(\mathbf{1} - \mathbf{f}_{u,q,t}) \quad \forall u \in \mathcal{U}, q \in \mathcal{Q}, t \in \mathcal{T}_f, \quad (16)$$

$$\mathbf{1} \cdot \mathbf{f}_{u,q,t} \geq 1 \quad \forall u \in \mathcal{U}, q \in \mathcal{Q}, t \in \mathcal{T}. \quad (17)$$

Wind zones are incorporated similarly by using a set $\mathcal{P}$ of hyperplanes $\tilde{N}_p$ for $p \in \mathcal{P}$ of the form $\{\mathbf{x} \in \mathbb{R}^2 | \mathbf{B}_{p,t}\mathbf{x} \leq \hat{\mathbf{c}}_{p,t}\}$ with $\mathbf{B}_{p,t} \in \mathbb{R}^{\tilde{N}_p \times 2}$ and $\mathbf{c}_{p,t} \in \mathbb{R}^{\tilde{N}_p}$. Utilizing wind vector $\mathbf{w}_{p,t} = (w_{p,t}^x, w_{p,t}^y) \in \mathbb{R}^2$ that describes the wind velocity per wind zone $p \in \mathcal{P}$ at every coarse time step $t \in \mathcal{T}$ and by defining binary decision variables $\hat{\mathbf{w}}_{u,p,t} = (\hat{w}_{u,p,t}^1, \dots, \hat{w}_{u,p,t}^{\tilde{N}_p})^T \in \{0, 1\}^{\tilde{N}_p}$ and $\omega_{u,p,t} \in \{0, 1\}$ for $u \in \mathcal{U}, p \in \mathcal{P}, t \in \mathcal{T}$ as follows

$$\hat{w}_{u,p,t}^k = \begin{cases} 1, & \text{if UAV } u \in \mathcal{U} \text{ is within wind zone } p \in \mathcal{P} \\ & \text{w.r.t. hyperplane } k \text{ in } \{1, \dots, \tilde{N}_p\} \text{ at time step } t \in \mathcal{T}, \\ 0, & \text{else,} \end{cases} \quad (18)$$

$$\omega_{u,p,t} = \begin{cases} 1, & \text{if UAV } u \in \mathcal{U} \text{ is affected by wind zone } p \in \mathcal{P} \text{ at time} \\ & \text{step } t \in \mathcal{T}, \\ 0, & \text{else,} \end{cases} \quad (19)$$

we are able to map all decisions from the $\tilde{N}_p$ hyperplanes to a single decision (from $\hat{\mathbf{w}}_{u,p,t}$ to $\omega_{u,p,t}$):

$$\mathbf{B}_{q,\hat{t}} \mathbf{r}_{u,t} \geq \hat{\mathbf{c}}_{p,t} - M \hat{\mathbf{w}}_{u,p,t} \quad \forall u \in \mathcal{U}, p \in \mathcal{P}, t \in \mathcal{T}, \quad (20)$$

$$\mathbf{1}^T \cdot \hat{\mathbf{w}}_{u,p,t} \leq \tilde{N}_p + \omega_{u,p,t} - 1 \quad \forall u \in \mathcal{U}, p \in \mathcal{P}, t \in \mathcal{T}, \quad (21)$$

$$\mathbf{1}^T \cdot \hat{\mathbf{w}}_{u,p,t} \geq \tilde{N}_p \omega_{u,p,t} \quad \forall u \in \mathcal{U}, p \in \mathcal{P}, t \in \mathcal{T}. \quad (22)$$

Because this approach causes a discretization error due to the two different time discretizations, we can reduce this effect using a convex combination of wind vectors $\mathbf{w}_{p,t}$ at the fine steps between the coarse ones with convex combination parameter $\mu = \frac{t \bmod n_f}{n_f}$ and single wind zone decision variables $\omega_{u,p,t}$:

$$\tilde{\mathbf{w}}_{u,t} := \sum_{p \in \mathcal{P}} (1 - \mu) \omega_{u,p,\hat{t}} \mathbf{w}_{p,\hat{t}} + \mu \omega_{u,p,\hat{t}+n_f} \mathbf{w}_{p,\hat{t}+n_f} \quad (23)$$

$$\forall u \in \mathcal{U}, t \in \mathcal{T}_f \setminus \{n_f T\}$$

Each waypoint $w \in \mathcal{W}$ with its respective location $\mathbf{p}_w \in \mathbb{R}^2$ can be visited only within time window $\mathcal{T}_w \subseteq \mathcal{T}$ by a UAV entering a waypoint's predefined operational range $\delta_{u,w} \in \mathbb{R}_{>0}$, $u \in \mathcal{U}$, $w \in \mathcal{W}$. A waypoint $w \in \mathcal{W}_u$ is considered collected when UAV $u \in \mathcal{U}$ is within its maximum operational distance $\delta_{u,w} \in \mathbb{R}_{>0}$. So we introduce waypoint binary variables

$$d_{u,w,t} = \begin{cases} 1, & \text{if UAV } u \in \mathcal{U} \text{ is within operational range } \delta_{u,w} \\ & \text{of waypoint } w \in \mathcal{W}_u \text{ at time step } t \in \mathcal{T}_w, \\ 0, & \text{else,} \end{cases} \quad (24)$$

and constraints

$$\|\mathbf{r}_{u,t} - \mathbf{p}_w\|_2 \leq \delta_{u,w} + M(1 - d_{u,w,t}) \quad \forall u \in \mathcal{U}, w \in \mathcal{W}, t \in \mathcal{T}_w, \quad (25)$$

$$\sum_{u \in \mathcal{U}, t \in \mathcal{T}_w} d_{u,w,t} \leq 1 \quad \forall w \in \mathcal{W}, \quad (26)$$

where M is a sufficiently large constant and $\mathbf{r}_{u,t}$, $\mathbf{p}_w$ are the UAV's and the waypoint's position, respectively. Eqs. (25) ensure that a waypoint $w \in \mathcal{W}$ only counts as collect, when the UAV $u \in \mathcal{U}$ is inside its operational range $\delta_{u,w}$ and Eqs. (26) ensure that every waypoint is visited at most once. To describe the flight dynamics of the UAVs, we use Newton's equation of motion

$$\mathbf{a} = \dot{\mathbf{v}} = \frac{d\mathbf{v}}{dt}, \quad \mathbf{v} = \dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt}, \quad (27)$$

for position $\mathbf{r}$, velocity $\mathbf{v}$ and acceleration $\mathbf{a}$. We define velocity $\mathbf{v}_{u,t} \in \mathbb{R}^2$ and acceleration $\mathbf{a}_{u,t} \in \mathbb{R}^2$ of UAV $u \in \mathcal{U}$ at time step $t \in \mathcal{T}_f$ and restrict it from above by

maximum velocity $v_u^{max} \in \mathbb{R}_{>0}$ and maximum acceleration $a_u^{max} \in \mathbb{R}_{>0}$, respectively:

$$\|\mathbf{v}_{u,t}\|_2 \leq v_u^{max} b_{u,t} \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f, \quad (28)$$

$$\|\mathbf{a}_{u,t}\|_2 \leq a_u^{max} b_{u,t} \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f. \quad (29)$$

We discretize Eq. (27) in t using the n -th order Taylor expansion

$$\begin{aligned} f(x) &= f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots \\ &= \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \approx \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k =: T_f^n(x, x_0). \end{aligned} \quad (30)$$

Here, we choose $n = 2$ and update the position of the UAVs $\mathbf{r}_{u,t} = (r_{u,t}^x, r_{u,t}^y)$ for the next time step $t + 1$ via

$$\mathbf{r}_{u,t+1} = \mathbf{r}_{u,t} + \Delta t_f \dot{\mathbf{r}}_{u,t} + \frac{(\Delta t_f)^2}{2} \ddot{\mathbf{r}}_{u,t} \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f \setminus \{n_f T\}. \quad (31)$$

Utilizing velocity $\mathbf{v}_{u,t}$, acceleration $\mathbf{a}_{u,t}$, and convex combination wind vector $\tilde{\mathbf{w}}_{u,t}$ we find

$$\dot{\mathbf{r}}_{u,t} = \mathbf{v}_{u,t} + \tilde{\mathbf{w}}_{u,\hat{t}}, \quad \ddot{\mathbf{r}}_{u,t} = \mathbf{a}_{u,t} \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f \setminus \{n_f T\}. \quad (32)$$

Inserting Eq. (32) in Eq. (31) and applying first order Taylor expansion of $\mathbf{v}_{u,t+1}$ we obtain the following equations to update the position $\mathbf{r}_{u,t}$ and the velocity $\mathbf{v}_{u,t}$ of the UAVs for the next time step $t + 1 \in \mathcal{T}_f$:

$$\mathbf{r}_{u,t+1} = \mathbf{r}_{u,t} + \Delta t_f (\mathbf{v}_{u,t} + \tilde{\mathbf{w}}_{u,\hat{t}}) + \frac{(\Delta t_f)^2}{2} \mathbf{a}_{u,t} \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f \setminus \{n_f T\}, \quad (33)$$

$$\mathbf{v}_{u,t+1} = \mathbf{v}_{u,t} + \Delta t_f \mathbf{a}_{u,t} \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f \setminus \{n_f T\}. \quad (34)$$

To consider the battery level of a UAV, a safety energy level $g_u^{\min} \in [0, 1)$ is required. We introduce a parameter $n_{u,j} \in [0, 1)$ for the battery consumption of UAV $u \in \mathcal{U}$ depending on its velocity $j \in \mathcal{V}_u$, where $\mathcal{V}_u$ is a velocity band that splits the velocity range $[0, v_u^{max}]$ using $\theta_{u,1} = 0 < \theta_{u,2} < \dots < \theta_{u,|\mathcal{V}_u|} = v_u^{max}$. The remaining battery level of a UAV at a certain timestep is defined by variable $g_{u,t} \in [g_u^{\min}, 1]$, $u \in \mathcal{U}$, $t \in \mathcal{T}_f$ that shall not exceed a minimum battery level of $g_u^{\min}$. We introduce the binary variables

$$s_{u,j,t} = \begin{cases} 1, & \text{if UAV } u \in \mathcal{U} \text{ is in velocity band } j \in \mathcal{V}_u \\ & \text{at time step } t \in \mathcal{T}, \\ 0, & \text{else.} \end{cases} \quad (35)$$

The remaining battery power $g_{u,t}$ at timestep $t \in \mathcal{T}_f$ starts at battery level $g_u(0) = g_u^{start}$ and is described by:

$$g_{u,t+1} = g_{u,t} - \Delta t_f \left(\sum_{j \in \mathcal{V}_u} n_{u,j} s_{u,j,\hat{t}} \right) \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f \setminus \{n_f T\}, \quad (36)$$

$$\sum_{j \in \mathcal{V}_u} s_{u,j,t} = b_{u,t} \quad \forall u \in \mathcal{U}, t \in \mathcal{T}, \quad (37)$$

$$\sum_{j \in \mathcal{V}_u} \theta_{u,j} s_{u,j,t} \leq \|\mathbf{v}_{u,t}\|_2 \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f, \quad (38)$$

$$g_{u,t} \geq g_u^{\min} \quad \forall u \in \mathcal{U}, t \in \mathcal{T}_f. \quad (39)$$

Eqs. (36) describe the battery consumption per timestep airborne (with step size Δt_f), depending on the battery consumption $n_{u,j}$ per current velocity band determined by $s_{u,j,t}$. Eqs. (37) ensure a velocity band is only selected when the UAV is midair, as indicated by binaries $b_{u,t}$, and Eqs. (38) couple the current velocity band $\theta_{u,j}$ to the UAV's current velocity $\mathbf{v}_{u,t}$. Eq. 39 ensures complying with the minimum battery level $g_u^{\min}$ at every time step.

By introducing score values $S_w \in \mathbb{R}_{>0}$, to indicate the importance of every waypoint $w \in \mathcal{W}$, our objective is now to collect the waypoints with a high score value as early as possible. We also want the operation to proceed as quickly as possible. To measure the total duration of the operation, we introduce a decision variable $\theta^{\max} \in \mathbb{R}_+$ that denotes the mission duration and is determined by the last time step $t \in \mathcal{T}$ a UAV $u \in \mathcal{U}$ is airborne (as represented by airborne decision variable $b_{u,t}$) as guaranteed by the constraints

$$t b_{u,t} \leq \theta^{\max} \quad \forall u \in \mathcal{U}, t \in \mathcal{T}. \quad (40)$$

Finally, the objective function takes the form

$$\max \sum_{u \in \mathcal{U}, w \in \mathcal{W}, t \in \mathcal{T}_\omega} \frac{S_w d_{u,w,t}}{t+1} - \frac{\theta^{\max}}{M^{\text{obj}}}, \quad (41)$$

maximizing the with S_w weighted collected waypoints (as indicated by binary variables $d_{u,w,t}$) while minimizing the mission duration $\theta^{\max}$. Here $M^{\text{obj}} \in \mathbb{R}_{>0}$ is a scaling coefficient which, with a suitable choice, ensures that a shorter mission duration is not more important than collecting a waypoint. Note that instead of term $t+1$, also $(t+1)^n$, $n > 1$ can be used in the objective function to further increase the effect that strongly weighted waypoints are visited first. The choice of M^{obj} controls the trade-off between maximizing the waypoint coverage and minimizing mission duration. If M^{obj} is chosen too large, the waypoint reward term $\sum \frac{S_w d_{u,w,t}}{t+1}$ in Eq. (41) dominates the objective. In this case, the optimizer prioritizes visiting all reachable waypoints as early as possible, neglecting shortening of the overall flight time. Consequently, once all waypoint rewards are collected, the UAV may exhibit unrealistic behavior, such as loitering or drifting slowly instead of directly returning to the end point, since reducing the remaining flight time has only marginal impact on the objective value. Conversely, if M^{obj} is chosen too small, minimizing mission duration becomes overly dominant, and the optimizer may prefer shorter trajectories skipping waypoints even when they could be feasibly visited. In practice, M^{obj} must therefore be selected within a moderate range to balance ensuring realistic trajectories that visit every relevant waypoint possible while still encouraging fast mission completion.

Finally, for this model formulation to become a MILP, we need to linearize the Euclidean norm $\|\cdot\|_2$ in Eqs. (3), (25), (28), (29) and (38), approximating it by a linear combination of the 1-norm and uniform norm (sup norm) as detailed by Fügenschuh et al. (2019).

4 UAV Hangar Location Optimization

Classical EMS location models, such as the *Maximum Covering Location Problem (MCLP)* (Church and ReVelle 1974) or *Maximum Expected Covering Location Problem (MEXCLP)* (Daskin 1983), etc. focus on availability aspects to ensure coverage even when some EMS vehicles are engaged on duty. These models are very well studied, however, they typically consider individual coverage, as a single vehicle typically is assigned to one request each. This may suffice for standard EMS systems. In our use case where we need to search a large area, the deployment of multiple cooperating vehicles, in this case UAVs, offers significant advantages over an individual mission. To model this, Berman et al. (2011) developed a *cooperative covering location* approach, allowing the cooperative use of rescue resources. Additionally, we incorporate resource selection and sizing, considering different facility and vehicle types that may fulfill various tasks and assignments. This is particularly useful in cases where demand requires diverse services, for example a patient may require primary medical care from qualified first responders, assistance from an emergency doctor, transportation, medication or auxiliary devices like Automated External Defibrillators (AEDs). In our case, the UAVs might fulfill different tasks to help the distressed swimmer, e.g. detection, surveillance or delivery of flotation devices. We present the mathematical formulation of the MILP model below.

4.1 Model Framework

In the following, we present some slightly different MILP model formulations that represent cooperative maximum covering approaches for UAV hangar and UAV swarm sizing and facility location optimization. This model is based on the Cooperative Maximum Covering Location Problem (CMCLP) introduced by Berman et al. (2011) and incorporates hangars and UAVs of different types, sizes and purposes as well as joint UAV operations.

We consider a two-dimensional polygonal area of interest $\mathcal{A}$. Next, we define a discrete set of potential facility locations or UAV hangar sites $\mathcal{F} \neq \emptyset$ with $\mathcal{F} \subset \mathcal{A}$. The maximum capacity of a facility $i \in \mathcal{F}$ is denoted as $F_i^{max} \in \mathbb{R}_+$ and indicates the maximum hangar size that is suitable for this location. This can be particularly important if there are space limitations at the site or the ground is unstable. We consider a discrete set of UAV types $\mathcal{U}$ and a discrete set of UAV hangar types $\mathcal{H}$. We incorporate different purposes of the servers, here the UAVs, by introducing a discrete set of coverage contribution types $\mathcal{C}$ and a discrete set of demand points $\mathcal{D}$ with corresponding demand weights $d_{jc} \in \mathbb{R}_+$, $j \in \mathcal{D}$, $c \in \mathcal{C}$, determining how much demand of type c emerges at demand point j . This allows us to consider multiple

UAVs of different types that serve the same or a different purpose for the mission (e.g. searching, surveillance, rescue, delivery) because of a UAV's different characteristics (rescue equipment, camera sensors, battery capacity, payload, etc.). We introduce vehicle sizes $\psi_u \in \mathbb{R}_+$, a maximum number of available vehicles $N_u^{\max} \in \mathbb{N}$ for $u \in \mathcal{U}$, hangar sizes $s_h \in \mathbb{R}_+$ and a maximum number of available hangars $P_h^{\max} \in \mathbb{N}$ for $h \in \mathcal{H}$. Parameter $p \in \mathbb{N}$ denotes the number of UAV hangars to be set up. The contribution matrix $\mathcal{L} = (l_{uc}) \in [0, 1]^{|\mathcal{U}| \times |\mathcal{C}|}$ defines the contribution of a UAV $u \in \mathcal{U}$ to contribution type $c \in \mathcal{C}$ and can be interpreted as a percentage.

To incorporate a server cooperation aspect, we apply the concept of Berman et al. (2011), introducing a non-negative and non-increasing so-called *signal function* $\phi(\delta)$ and *aggregation operator* Φ of distance $\delta \in \mathbb{R}_+$. This signal function can be interpreted as a signal that becomes weaker with increasing distance. The aggregation operator then defines how signals from different signal emitters amplify each other. In our case, we utilize the *summation aggregation operator*, in general denoted as $\Phi_\Sigma = \sum_{i \in \mathcal{F}} x_i \phi(\delta_{ij})$, where x_i represents the number of UAVs at site $i \in \mathcal{F}$ and $\delta_{ij} \in \mathbb{R}_+$ is a distance from potential facility i to demand point j . We limit our consideration to Φ_Σ but there are other aggregation operators, such as polynomial models, the exponential model, the Fermi-type model, the Elfes model or the cubic model (Karatas 2017).

We define distance δ in terms of time, by introducing flight time $t_{uij} \in \mathbb{R}_+$ that describes the take-off from potential facility site $i \in \mathcal{F}$ to operation height $v \in \mathbb{R}_{>0}$ and approaching demand point $j \in \mathcal{D}$. We consider UAV hangar-specific delay τ_h , $h \in \mathcal{H}$ that indicates how long it takes until a UAV can take off from the hangar, e.g. until a closed hangar has opened its doors. Introducing a UAV's battery capacity $\sigma_u \in \mathbb{R}_+$ we define signal function $\phi(t_{uij}, \tau_h, \sigma_u)$ as the remaining time available to perform the search in the search area at demand point j after approaching it from potential facility site i from hangar h with UAV u . We also consider the return flight after the mission and this results in

$$\phi(t_{uij}, \tau_h, \sigma_u) = \max\{\sigma_u - 2t_{uij} - \tau_h, 0\} \quad \forall u \in \mathcal{U}, h \in \mathcal{H}, i \in \mathcal{F}, j \in \mathcal{D}. \quad (42)$$

In the following, we use the notation $\phi_{uijh} = \phi(t_{uij}, \tau_h, \sigma_u)$ to clarify that these function values are calculated in advance and used as parameters in the model. We incorporate *battery-wise coverage* by defining battery thresholds $T_{jc} \in \mathbb{R}_+$ for all contribution types $c \in \mathcal{C}$ and demand points $j \in \mathcal{D}$. When the aggregated signal exceeds the threshold, demand point j is covered. Additionally, we define a minimum battery threshold $T_{jc}^{\min} < T_{jc}$ which needs to be exceeded to guarantee at least a minimum coverage for every demand point. As our aim is not just to ensure coverage battery-wise but also to approach the scene as quickly as possible, we need to incorporate *time-wise coverage* as well. So we define time thresholds $\hat{T}_{jc}^{\min}, \hat{T}_{jc} \in \mathbb{R}_+$ with $\hat{T}_{jc}^{\min} > \hat{T}_{jc}$ and time coverage tensor $\mathbf{C} = (C_{cuijh})$ for $c \in \mathcal{C}$, $u \in \mathcal{U}$, $i \in \mathcal{F}$, $j \in \mathcal{D}$ and $h \in \mathcal{H}$ with

$$C_{cuijh} = \begin{cases} 1, & \text{if } l_{uc}(\tau_h + t_{uij}) \leq \hat{T}_{jc} \\ 0, & \text{else,} \end{cases} \quad (43)$$

and minimum time coverage tensor $C^{\min} = (C_{cuijh}^{\min})$ with

$$C_{cuijh}^{\min} = \begin{cases} 1, & \text{if } l_{uc}(\tau_h + t_{uij}) \leq \hat{T}_{jc}^{\min} \\ 0, & \text{else.} \end{cases} \quad (44)$$

Next, we define decision variables $n_{uih} \in \mathbb{N} \cup \{0\}$ that denote the number of UAVs of type $u \in \mathcal{U}$ stored in hangar of type $h \in \mathcal{H}$ located at site $i \in \mathcal{F}$. To determine if a hangar of type $h \in \mathcal{H}$ is located at site $i \in \mathcal{F}$, we define binary variables

$$a_{ih} = \begin{cases} 1, & \text{if facility } i \in \mathcal{F} \text{ of type } h \in \mathcal{H} \text{ is open,} \\ 0, & \text{else.} \end{cases} \quad (45)$$

We incorporate the coverage mechanism by defining covering decision variables

$$y_{jc} = \begin{cases} 1, & \text{if demand point } j \in \mathcal{D} \text{ is covered for contribution} \\ & \text{type } c \in \mathcal{C} \text{ w.r.t. } T_{jc}, \\ 0, & \text{else,} \end{cases} \quad (46)$$

$$y_{jc}^{\min} = \begin{cases} 1, & \text{if demand point } j \in \mathcal{D} \text{ is covered for contribution} \\ & \text{type } c \in \mathcal{C} \text{ w.r.t. } T_{jc}^{\min}, \\ 0, & \text{else,} \end{cases} \quad (47)$$

$$\hat{y}_{jc} = \begin{cases} 1, & \text{if demand point } j \in \mathcal{D} \text{ is covered for contribution} \\ & \text{type } c \in \mathcal{C} \text{ w.r.t. } \hat{T}_{jc}, \\ 0, & \text{else,} \end{cases} \quad (48)$$

and

$$\hat{y}_{jc}^{\min} = \begin{cases} 1, & \text{if demand point } j \in \mathcal{D} \text{ is covered for contribution} \\ & \text{type } c \in \mathcal{C} \text{ w.r.t. } \hat{T}_{jc}^{\min}, \\ 0, & \text{else.} \end{cases} \quad (49)$$

In the following we formulate the model constraints. Firstly, we want to locate at most $p \in \mathbb{N}$ hangars:

$$\sum_{h \in \mathcal{H}, i \in \mathcal{F}} a_{ih} \leq p. \quad (50)$$

The number of facilities to be opened or hangars to be located are restricted by the maximum number of available hangars $P_h^{\max} \in \mathbb{N}$,

$$\sum_{i \in \mathcal{F}} a_{ih} \leq P_h^{\max} \quad \forall h \in \mathcal{H}. \quad (51)$$

The number of UAVs to be allocated to a hangar is limited from above by the maximum number of available UAVs $N_u^{\max} \in \mathbb{N}$ of type $u \in \mathcal{U}$:

$$\sum_{i \in \mathcal{F}, h \in \mathcal{H}} n_{uih} \leq N_u^{\max} \quad \forall u \in \mathcal{U}. \quad (52)$$

We ensure to not exceed the capacity of a hangar $s_h \in \mathbb{R}_{>0}$ with the number of allocated UAVs:

$$\sum_{u \in \mathcal{U}} \psi_u n_{uih} \leq s_h a_{ih} \quad \forall i \in \mathcal{F}, h \in \mathcal{H}. \quad (53)$$

Additionally, we only allow placing at most one hangar at a potential facility site:

$$\sum_{h \in \mathcal{H}} a_{ih} \leq 1 \quad \forall i \in \mathcal{F}. \quad (54)$$

As some potential facility sites may be unsuitable for hangars that are too large or heavy, we introduce the size limit $F_i^{\max} \in \mathbb{R}_{>0}$ and constraint

$$\sum_{h \in \mathcal{H}} s_h a_{ih} \leq F_i^{\max} \quad \forall i \in \mathcal{F}. \quad (55)$$

We utilize the signal strength summation aggregation operator to determine the battery-wise covered demand points,

$$\sum_{u \in \mathcal{U}, h \in \mathcal{H}, i \in \mathcal{F}} n_{uih} l_{uc} \phi_{uijh} \geq T_{jc} y_{jc} \quad \forall j \in \mathcal{D}, c \in \mathcal{C}. \quad (56)$$

We also incorporate a minimum battery-wise coverage that needs to be guaranteed for every single demand point, in a strong formulation and a weak formulation, respectively.

$$\sum_{u \in \mathcal{U}, h \in \mathcal{H}, i \in \mathcal{F}} n_{uih} l_{uc} \phi_{uijh}^{\min} \geq T_{jc}^{\min} \quad \forall j \in \mathcal{D}, c \in \mathcal{C}, \quad (57a)$$

$$\sum_{u \in \mathcal{U}, h \in \mathcal{H}, i \in \mathcal{F}} n_{uih} l_{uc} \phi_{uijh}^{\min} \geq T_{jc}^{\min} y_{jc}^{\min} \quad \forall j \in \mathcal{D}, c \in \mathcal{C}. \quad (57b)$$

Since the strong formulation (57a) is very strict and leads to infeasible instances as soon as even a single point cannot be covered, this formulation is relaxed to (57b). Weak minimum battery-wise coverage can now be incorporated by adding a penalty term in the objective function. Equivalently, to ensure time-wise coverage, we add the constraint

$$\sum_{u \in \mathcal{U}, i \in \mathcal{F}, h \in \mathcal{H}} n_{uih} C_{uijh} \geq \hat{y}_{jc} \quad \forall j \in \mathcal{D}, c \in \mathcal{C}, \quad (58)$$

and incorporate strong and weak minimum time-wise coverage via

$$\sum_{u \in \mathcal{U}, i \in \mathcal{F}, h \in \mathcal{H}} n_{uih} C_{uijh}^{\min} \geq 1 \quad \forall j \in \mathcal{D}, c \in \mathcal{C}. \quad (59a)$$

$$\sum_{u \in \mathcal{U}, i \in \mathcal{F}, h \in \mathcal{H}} n_{uih} C_{cuijh}^{\min} \geq \hat{y}_{jc}^{\min} \quad \forall j \in \mathcal{D}, c \in \mathcal{C}. \quad (59b)$$

All the Eqs. (56) – (59b) follow the same principal: The y decision variables can only reach a value equal to one if the coverage summed over the assigned servers exceeds a certain threshold value.

Now that the model constraints are defined, we can specify objective functions for the different approaches, namely the models without minimum coverage, with strict minimum coverage and with weak minimum coverage. We name the first model without minimum coverage the *Capacitated Cooperative Maximum Covering Location Problem (CCMCLP)* and define the following objective which aims to maximize the weighted covered demand points w.r.t. battery and time coverage. The objective function is defined by

$$\max \sum_{c, u, i, j, h} d_{jc} n_{uih} y_{jc} \hat{y}_{jc} - \frac{1}{K} \sum_{c, u, i, j, h} (t_{uij} + \tau_h) n_{uih} y_{jc} \hat{y}_{jc}, \quad (60)$$

where $K \in \mathbb{R}_{>0}$ is a scaling constant. Eq. (60) is non-linear because of the product of decision variables $n_{uih} y_{jc} \hat{y}_{jc}$. Hence, we linearize this term, introducing variables $\tilde{z}_{cuijh} \in \mathbb{R}$ which will represent the product above. We utilize a big-M approach, defining a sufficiently large constant $M^{\text{loc}} \in \mathbb{R}$ and the following constraints:

$$\tilde{z}_{cuijh} \leq n_{uih}, \quad (61) \quad \tilde{z}_{cuijh} \leq M^{\text{loc}} y_{jc}, \quad (62)$$

$$\tilde{z}_{cuijh} \leq M^{\text{loc}} \hat{y}_{jc} \quad \forall c \in \mathcal{C}, u \in \mathcal{U}, i \in \mathcal{F}, j \in \mathcal{D}, h \in \mathcal{H}. \quad (63)$$

Note that this guarantees that $\tilde{z}_{cuijh}$ can only equal n_{uih} , if both binaries y_{jc} and $\hat{y}_{jc}$ are equal to one. The linearized objective function can be written as

$$\max \sum_{c, u, i, j, h} \left(d_{jc} - \frac{1}{K} (t_{uij} + \tau_h) \right) \tilde{z}_{cuijh}. \quad (64)$$

The CCMCLP is defined by the Eqs. (50) – (56), (58), (61) – (64). The second model variant that includes strict minimum coverage is named *Capacitated Cooperative Maximum Guaranteed Minimum Covering Location Problem (CCMGMCLP)*. It is defined by the Eqs. (50) – (57a), (58), (59a), (61) – (64). The third model variant with weak minimum coverage is named *Capacitated Cooperative Maximum Weak Minimum Covering Location Problem (CCMWMCLP)*. Analogous to the above approach we introduce real variables $\tilde{z}_{cuijh}^{\min}$ and constraints

$$\tilde{z}_{cuijh}^{\min} \leq n_{uih}, \quad (65) \quad \tilde{z}_{cuijh}^{\min} \leq M^{\text{loc}} y_{jc}^{\min}, \quad (66)$$

$$\tilde{z}_{cuijh}^{\min} \leq M^{\text{loc}} \hat{y}_{jc}^{\min} \quad \forall c \in \mathcal{C}, u \in \mathcal{U}, i \in \mathcal{F}, j \in \mathcal{D}, h \in \mathcal{H}. \quad (67)$$

as well as objective function

$$\begin{aligned}
\max \quad & \lambda_1 \sum_{c,u,i,j,h} d_{jc} \tilde{z}_{cuijh} - \frac{1}{K_1} \sum_{c,u,i,j,h} (t_{uij} + \tau_h) \tilde{z}_{cuijh} \\
& + \lambda_2 \sum_{c,u,i,j,h} d_{jc} \tilde{z}_{cuijh}^{\min} - \frac{1}{K_2} \sum_{c,u,i,j,h} (t_{uij} + \tau_h) \tilde{z}_{cuijh}^{\min},
\end{aligned} \tag{68}$$

with parameters $\lambda_1, \lambda_2 \in [0, 1]$, $\lambda_1 + \lambda_2 = 1$ and $K_1, K_2 \in \mathbb{R}_{>0}$. The CCMWMCLP is defined by the Eqs. (50) – (56), (57b), (58), (59b), (61) – (63), (65) – (67), (68). Since the variables $\tilde{z}_{cuijh}, \tilde{z}_{cuijh}^{\min}$ are to represent the product of $n_{uih}, y_{jc}, \hat{y}_{jc}$ and $n_{uih}, y_{jc}^{\min}, \hat{y}_{jc}^{\min}$, respectively, which are limited by $N_u^{\max}$ by Eq. (52), we can compute the big-M constant M^{loc} by

$$M^{\text{loc}} = \max_{u \in \mathcal{U}} N_u^{\max}. \tag{69}$$

This choice yields a tight Big-M value. In particular, the variables $\tilde{z}_{cuijh}$ and $\tilde{z}_{cuijh}^{\min}$ represent the product of binary variables and integer variable n_{uih} , which is explicitly bounded by $N_u^{\max}$. Hence, M^{loc} corresponds to the smallest valid upper bound on $\tilde{z}_{cuijh}$ and $\tilde{z}_{cuijh}^{\min}$ and does not weaken the formulation. Since the number of available UAVs $N_u^{\max}$ is naturally limited in practical applications, the resulting Big-M values remain moderate and do not lead to numerical stability issues.

5 Case Study and Scenario Analysis

In the following, the models from Sections 3 and 4 are applied to the *RescueFly* test area located within the *Lusatian Lake District*, which is located partially in Brandenburg and partially in the Free State of Saxony in Eastern Germany. This is an open-cast lignite mining area in which some sites are still in operation, while others that are no longer in use have been flooded and converted into artificial lakes.

5.1 Location Optimization

To apply the MILP from Section 4, the associated parameters must first be calculated. First, we need to determine demand points $j \in \mathcal{D}$ and corresponding demand weights d_{jc} . These weights represent the frequency of use of the swimming areas at the lakes. To depict this reasonably, we gather data from various sources and assign appropriate weights. First, we define the area of interest $\mathcal{A}$ as a part of the Lusatian Lake District which includes Lake Partwitz, Lake Geierswalde, Lake Sedlitz and Lake Senftenberg by specifying its coordinates in World Geodetic System 1984 (WGS84) format and using OpenStreetMap (OSM) to access relevant features within $\mathcal{A}$ as listed in Table 1 and illustrated in Figure 3a.

We also consider region-specific negative influencing factors: When flooding inactive lignite mining pits, landslides are possible even years after the flooding finished. We access the landslide-prone areas identified by authorities (Lausitzer und Mitteldeutsche Bergbau-Verwaltungsgesellschaft mbH (LMBV)) via an Application Programming Interface (API) (Lausitzer und Mitteldeutsche Bergbau-

Table 1 OSM map features related to the use of bathing areas

Key	Value	Description
natural	water	Inland water body (lake, pond, canal, etc.)
	sand	Area covered with sand
	beach	land form along water consisting of sand, shingle or other loose material
leisure amenity	swimming_area	An area for swimming (marked by rope, buoys, etc.)
	parking	Car park
	surf_school	Surf school
tourism	camp_site	Camping site
	caravan_site	Caravan site
	meadow	A meadow or pasture (grass or non-woody plants)
landuse	recreation_ground	Open green space for general recreation
	residential	Residential place (house, apartment, flat, unit, etc.)
	forest	Managed forest or woodland plantation
sport	sailing	Sailing sport
	surfing	Surf sport
	swimming	Swimming sport
	water_polo	Water polo play field
	water_ski	Water ski area
	windsurfing	Windsurfing area

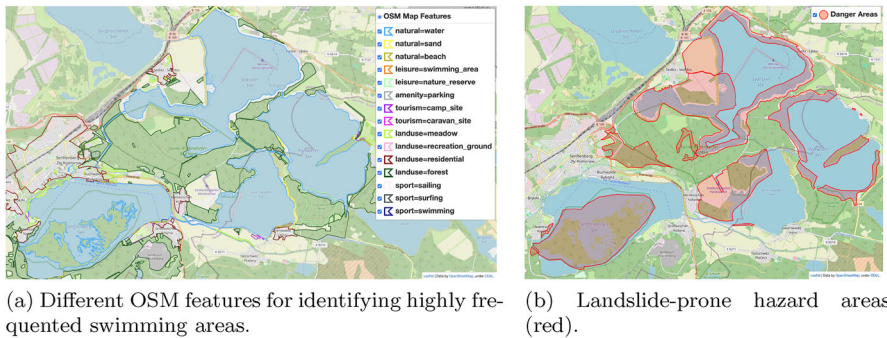


Fig. 3 OSM hotspot areas (left) and landslide-prone areas (right) within the Lusatian Lake District

Table 2 Utilized official feature names (German) of the dipul database (Digital Platform for Unmanned Aviation 2023) and corresponding translation (English)

Query feature (German)	Translation (English)
flugplaetze; flughafen; kontrollzonen; flugbeschaenkungsgebiete; bundesautobahnen; bundesstrassen; bahnanlagen; binnenwasserstrassen; seewasserstrassen; schiffahrtsanlagen; ; wohngrundstuecke; Freibaeuer; industrianlagen; kraftwerke; umspannwerke; stromleitungen; windkraftanlagen; justizvollzugsanstalten; militaerische_anlagen; labore; behoerden; diplomatische_vertretungen; internationale_organisationen; polizei; sicherheitsbehoerden; krankenhaeuser; nationalparks; naturschutzgebiete; ffh-gebiete; vogelschutzgebiete	Airfields; Airports; Control zones; Flight restriction areas; Federal highways; Federal roads; Railways; Inland waterways; Maritime waterways; Shipping facilities; Residential properties; Outdoor pools; Industrial facilities; Power plants; Substations; Power lines; Wind turbines; Correctional facilities; Military facilities; Laboratories; Authorities; Diplomatic representations; International organizations; Police; Security authorities; Hospitals; National parks; Nature reserves; Flora-fauna-habitat areas; Bird sanctuaries

Verwaltungsgesellschaft mbH 2024). They are depicted in Figure 3b. In addition, we collaborated with two local operations centers *Integrierte Regionalleitstelle Lausitz* and *Integrierte Regionalleitstelle Ostsachsen* to collect anonymized real-world historical data over a five-year period. These historical incidents are shown in Figure 4a. The Digital Platform for Unmanned Aviation (Digitale Plattform Unbemannte Luftfahrt (dipul)) provides official geographical zones according to German aviation regulations (§21h LuftVO) in which UAV operations are only possible under certain conditions (Federal Ministry of Transport and Digital Infrastructure 2026). They also offer an API (Digital Platform for Unmanned Aviation 2023) to display official restricted areas. Table 2 shows the selected features as visualized in Figure 4b.

5.1.1 Demand Points Localization

To identify demand point locations, we utilize expert knowledge about typical swimming areas, defined by 2D polygons. The corresponding discrete demand point of such a polygon is its centroid. The associated weight is then finally determined as the sum

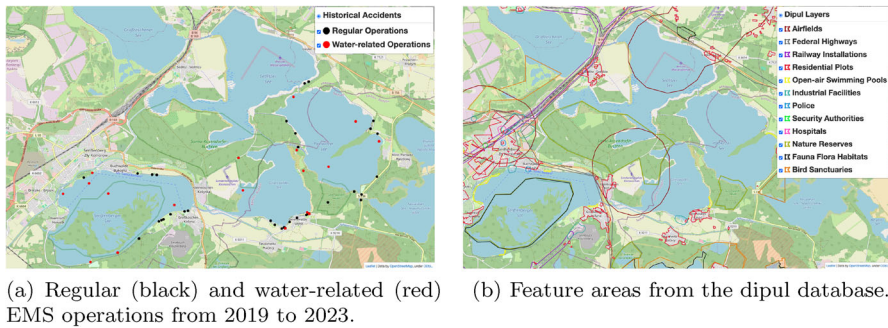


Fig. 4 Historical accidents (left) and dipul features (right) within the Lusatian Lake District

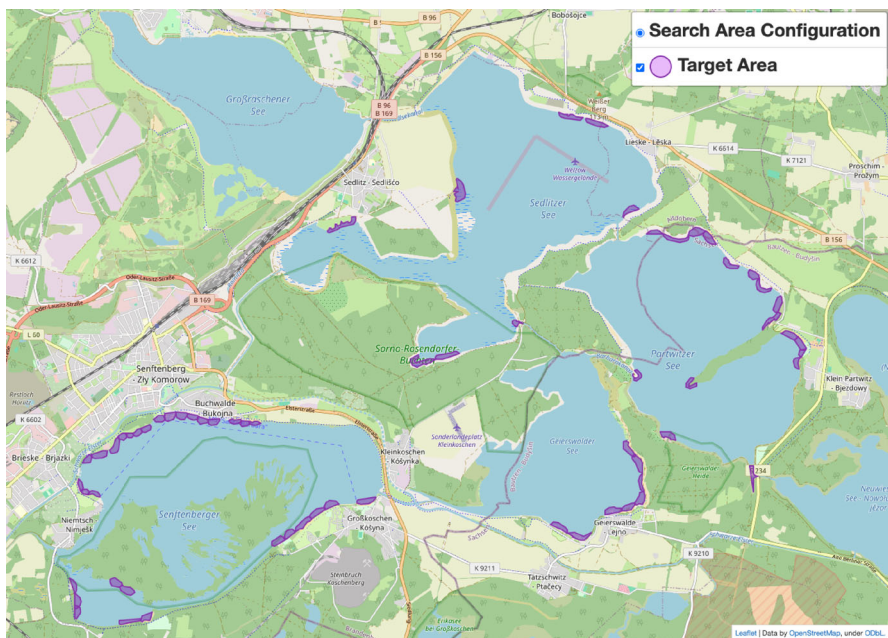


Fig. 5 The 65 highly frequented swimming areas (violet) defined with expert knowledge

of all weights of the equidistant points within the polygon. The corresponding target areas are visualized in Figure 5.

5.1.2 Demand Weighting

We determine demand weights $d_{jc} \geq 0$, by assigning individual positive or negative weighting for every demand point located within positively (e.g. beaches, camping sites, swimming areas) or negatively influencing areas (e.g. nature reserves, danger areas), respectively. Those influencing areas are mainly on land, so we introduce a buffer size to intersect the water area. This allows the demand points in the water to

Table 3 Demand weighting databases with corresponding features, buffer sizes in meters and weighting values

Database	Feature	Buffer [m]	Weight
OSM	swimming_area	300	1.0
	beach	200	1.0
	water	0	1.0
	swimming	500	0.5
	sand; camp_site, water_polo	300	0.5
	surf_school	500	0.4
	sailing	400	0.3
	caravan_site	300	0.3
	parking	500	0.2
	meadow; recreation_ground; residential; surfing	300	0.2
	water_ski, windsurfing	500	0.1
	nature_reserve	300	-0.3
Dipul	flora-fauna-habitats; bird sanctuaries	20.0	-0.2
	nature reserves	20.0	-0.3
LMBV		50	-0.5
Historical	water-related mission	400	1.0
	standard mission	400	0.5

be weighted according to weights listed in Table 3. The resulting demand is shown on a heat map in Figure 6.

5.1.3 Potential Facility Sites Localization

We employ an equidistant grid approach to determine the set of potential facility sites $\mathcal{F}$ that are exclusively located on land. We then exclude unsuitable sites utilizing layers shown in Table 4. Any of these layers make placing hangars very difficult to impossible. In a further pre-processing step, we remove redundant potential facility sites by introducing so-called *dominated* points:

Definition 1 (*Dominated point*) Consider distances $\delta_{ij} \in \mathbb{R}_+$ from potential facility site $i \in \mathcal{F}$ to demand point $j \in \mathcal{D}$. A point $i_1 \in \mathcal{F}$ is called dominated by $i_2 \in \mathcal{F}$, if it holds

$$d_{i_1 j} > d_{i_2 j} \forall j \in \mathcal{D}. \quad (70)$$

Of the original 940 potential facility sites, 106 dominated points were identified and removed. The effort is $\mathcal{O}(|\mathcal{F}| \cdot |\mathcal{D}|^2)$ and it took 5.7 seconds to identify all dominated points on a MacBook Pro (Intel Core i9 running 1 thread at 2.4 GHz clock speed and 64 GB RAM) using Python 3.8.0 (Python Software Foundation 2019). The resulting potential facility sites are depicted in Figure 7.

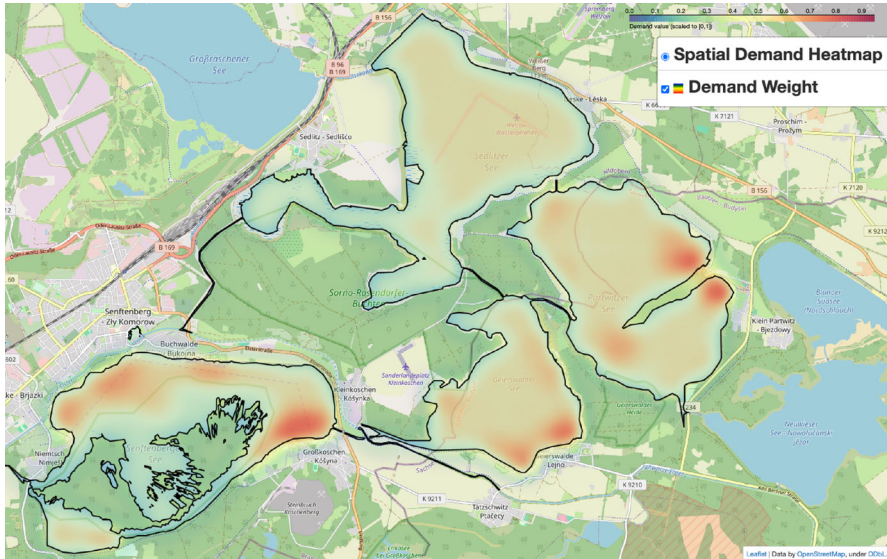


Fig. 6 Spatial demand visualized as a heat map indicating low (blue) to highly frequented areas (red) within the Lusatian Lake District

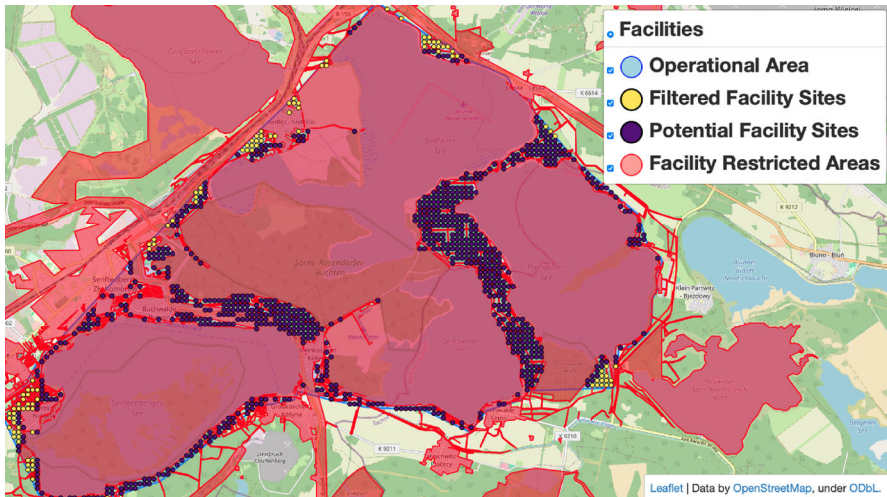


Fig. 7 Potential facility sites (violet), dominated points (yellow), and the corresponding exclusion zones (red) in the Lusatian Lake District

5.1.4 Distance Computation

We compute travel times t_{uij} of UAV u from potential site i to demand point j . This is the time it takes for a UAV $u \in \mathcal{U}$ to approach demand point $j \in \mathcal{D}$ from potential facility site $i \in \mathcal{F}$. Here, we have assumed straight, unconstrained point-to-point flights from i to j , resulting in a large number of pairwise travel-time estimations.

Table 4 Features for determining exclusion areas for potential facility sites

Database	Key	Value
OpenStreetMap	natural	water, beach
	leisure	swimming_area, pitch
	aeroway	aerodrome
	highway	motorway, trunk, primary, secondary, tertiary, unclassified, residential, motorway_link, trunk_link, primary_link, secondary_link, tertiary_link, living_street, service, pedestrian, track, bus_guideway, escape, raceway, road, busway, footway, bridleway, steps, corridor, path, via_ferrata
Dipul		airfields, airports, control zones, flight restriction areas, flight restriction areas, power plants, substations, power lines, wind turbines, correctional facilities, military facilities authorities, diplomatic representations, international organizations, national parks, nature reserves, flora-fauna-habitat areas, bird sanctuaries, residential properties, outdoor pools, railways, federal highways, federal roads
LMBV		

Embedding the full flight-planning MILP from Section 3 would require solving $|\mathcal{D}| \cdot |\mathcal{F}|$ optimization problems, which would substantially increase the computational burden although the problem size is significantly reduced by neglecting way points, restricted areas, and wind. For this reason, we use a simplified physics-based approach to approximate the flight times which remains consistent with the MILP flight planning model. We identified three different phases: (1) take-off from i to operation altitude v , (2) Uniform acceleration with rate $a_u^{\max}$, (3) Approach phase to j with maximum velocity $v_u^{\max}$. Utilizing climb rate c_u , we can compute the time spent in the first phase by $\frac{v}{c_u}$. The time spent in the second phase is computed by $\frac{v_u^{\max}}{a_u^{\max}}$. The distance covered in this phase can be computed by the *kinematic equation of motion*, describing the uniform acceleration for starting speed v_0 to target speed v_T by $v_T^2 = v_0^2 + 2ad$ where a is the acceleration and d is the distance traveled. Since $v_0 = 0$, $v_T = v_u^{\max}$, $a = a_u^{\max}$, we obtain $d = \frac{(v_u^{\max})^2}{2a_u^{\max}}$. Thus the distance that still has to be covered at top speed is equal to $\delta_{ij} - \frac{(v_u^{\max})^2}{2a_u^{\max}}$. We obtain

$$t_{uij} = \frac{v}{c_u} + \frac{v_u^{\max}}{a_u^{\max}} + \frac{\delta_{ij} - \frac{(v_u^{\max})^2}{2a_u^{\max}}}{v_u^{\max}} = \frac{v}{c_u} + \frac{v_u^{\max}}{2a_u^{\max}} + \frac{\delta_{ij}}{v_u^{\max}}. \quad (71)$$

If the distance δ_{ij} is so short that the UAV could not accelerate to $v_u^{\max}$, phase (3) is omitted. Thus, we obtain

$$t_{uij} = \begin{cases} \frac{v}{c_u} + \frac{v_u^{\max}}{2a_u^{\max}} + \frac{\delta_{ij}}{v_u^{\max}}, & \text{if } \frac{(v_u^{\max})^2}{2a_u^{\max}} > \delta_{ij}, \\ \frac{v}{c_u} + \sqrt{\frac{2\delta_{ij}}{a_u^{\max}}}, & \text{else.} \end{cases} \quad (72)$$

This simplified travel-time computation is consistent with the detailed two-dimensional flight planning MILP model in Section 3 because the corresponding two phases (2) and (3) described above are governed by the same physical limits on acceleration $a_u^{\max}$, and maximum velocity $v_u^{\max}$ that are also enforced in the MILP model. Hence, in the absence of wind and routing constraints, the detailed MILP model would yield flight times that are consistent with the approximation used here, with only minor deviations arising from the time discretization imposed by the fixed step size Δt . With this, we are finally set to compute the signal strengths via Eq. (42). Note that this approach does not allow for possible restriction areas to be included in the distance calculation. Since water rescue is a priority operation in which the overarching goal is to save a person's life and thus, there are typically no restricted areas, this simplification is acceptable for the large-scale location optimization. If travel times must explicitly account for restricted areas or waypoint routing, the flight planning model from Section 3 without waypoints can be utilized instead. As a trade-off, $|D| \cdot |F|$ small MILPs must be solved, leading to a substantially increased computational effort.

5.1.5 Model Solutions and Insights

First, we examine the case with one hangar type and one UAV type, which means $\mathcal{U} = \{0\}$, $\mathcal{H} = \{0\}$. These UAVs are responsible for localizing the swimmers, so the set of contribution types is defined by $\mathcal{C} = \{\text{"localization"}\}$ and the contribution matrix contains only one element: $\mathcal{L} = (l_{uc}) = (1)$. The parameters for solving the location problem using model variants *CCMCLP*, *CCMGMCLP* and *CCMWMCLP* from Section 4 are listed in Table 5. The instances were generated using the docplex package version 2.24.232 (International Business Machines Corporation 2022) and Python version 3.8.0 (Python Software Foundation 2019). The instances were solved to optimality using state-of-the-art commercial solver *IBM ILOG CPLEX* (International Business Machines Corporation 2023) version 21.1.1.0 on an Apple MacPro with an Intel Xeon W running 32 threads parallel at 3.20 GHz clock speed and 768 GB RAM with relative and absolute gap set to 0 and a time limit of 7200 seconds. The respective optimization results for the different models and different values of p are displayed in Figures 8 – 10. Note that the threshold values T_{jc} , $T_{jc}^{\min}$, $\hat{T}_{jc}$, $\hat{T}_{jc}^{\min}$ are quite sensitive and finding good-quality solutions requires a lot of fine-tuning. For example, if the threshold values are set too low and a few UAVs or hangars already fulfill the coverage and thus the objective function already reached its maximum value, the remaining UAVs are not placed at all, as no further improvement can be achieved. However, if the threshold values are defined too sharply, there are no feasible solutions for the *CCMGMCLP*. Figure 9 shows that for the parameter configuration selected here, feasible solutions only exist for $p \geq 6$. The *CCMWMCLP* was specifically developed

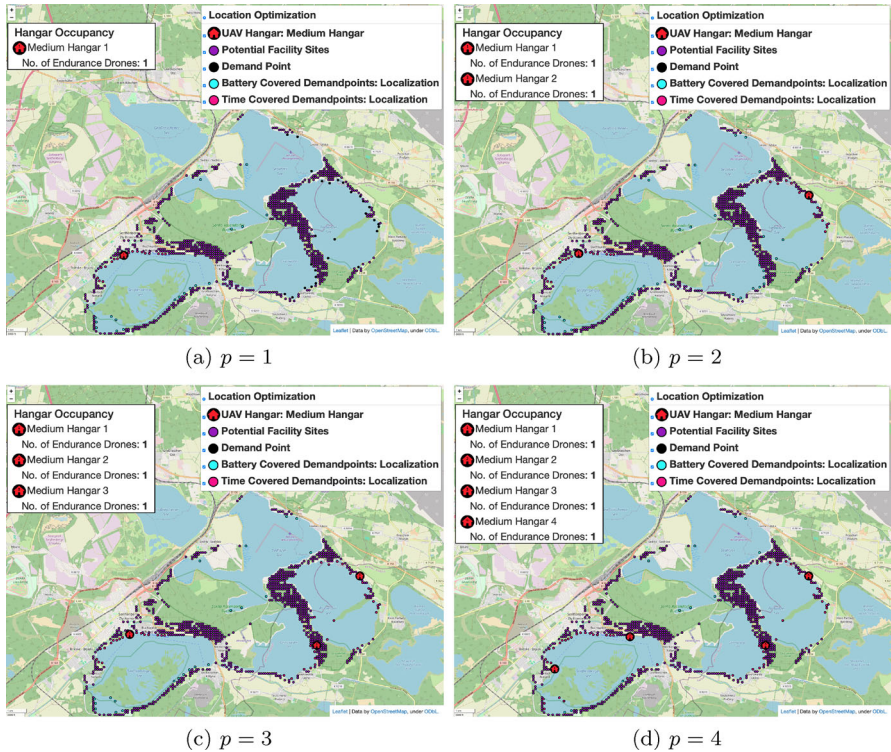


Fig. 8 CCMCLP model solution for the first use case with 1 to 4 hangars to be placed (from top left to bottom right)

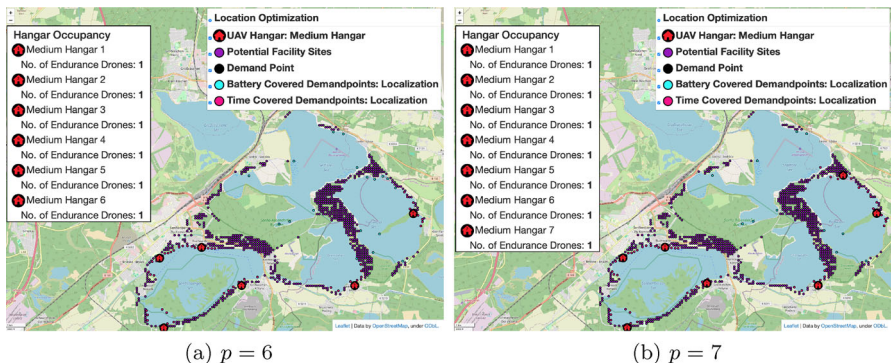


Fig. 9 CCMGMCLP model solution for the first use case with 6 and 7 hangars to be placed, respectively

to avoid these cases. Note that the solutions in Figure 9 guarantee a minimum time coverage of 120 seconds and minimum battery coverage of 1080 seconds for every single demand point, respectively. This does not hold for the *CCMWMCLP* solutions.

In the second use case, we consider three different UAV and hangar types, respectively. The hangars represent different models with varying capacity and the UAV types

Table 5 Location optimization model parameters for the first use case with only one UAV and hangar type

Name	Parameter	Value CCMCLP	CCMGMCPLP	CCMWMCLP
Maximum capacity of a potential facility site	$F_i^{\max}$	$3 \forall i \in \mathcal{F}$	$3 \forall i \in \mathcal{F}$	$3 \forall i \in \mathcal{F}$
UAV size	ψ_u	2	2	2
Available UAVs	$N_u^{\max}$	8	8	8
Hangar size	s_h	3	3	3
Available hangars	$P_h^{\max}$	8	8	8
Hangar delay	τ_h	8s	8s	8s
Operation altitude	v	50.0m	50.0m	50.0m
UAV battery capacity	σ_u	1800s	1800s	1800s
Battery threshold	T_{jc}	$1200s \forall j \in \mathcal{F}$	$1500s \forall j \in \mathcal{F}$	$1500s \forall j \in \mathcal{F}$
Minimum battery threshold	$T_{jc}^{\min}$	-	$1080s \forall j \in \mathcal{F}$	$1080s \forall j \in \mathcal{F}$
Time threshold	$\hat{T}_{jc}$	$100s \forall j \in \mathcal{F}$	$60s \forall j \in \mathcal{F}$	$60s \forall j \in \mathcal{F}$
Minimum time threshold	$\hat{T}_{jc}^{\min}$	-	$120s \forall j \in \mathcal{F}$	$120s \forall j \in \mathcal{F}$
UAV maximum speed	$v_u^{\max}$	$25.0 \frac{m}{s}$	$25.0 \frac{m}{s}$	$25.0 \frac{m}{s}$
UAV acceleration rate	$a_u^{\max}$	$5.0 \frac{m}{s^2}$	$5.0 \frac{m}{s^2}$	$5.0 \frac{m}{s^2}$
UAV climb rate	c_u	$5.0 \frac{m}{s^2}$	$5.0 \frac{m}{s^2}$	$5.0 \frac{m}{s^2}$

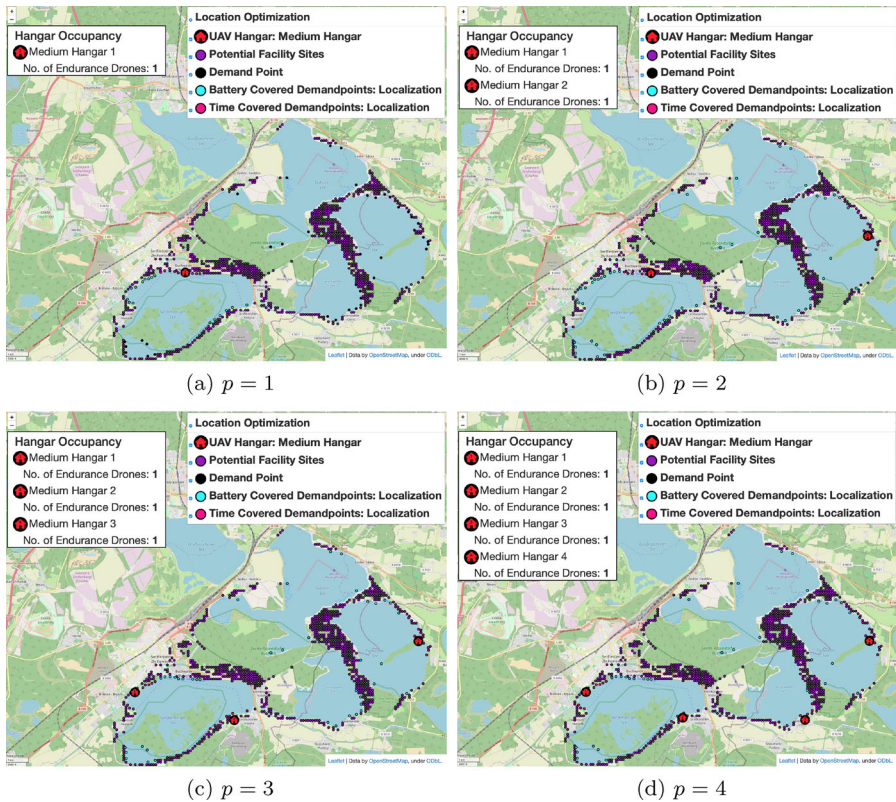
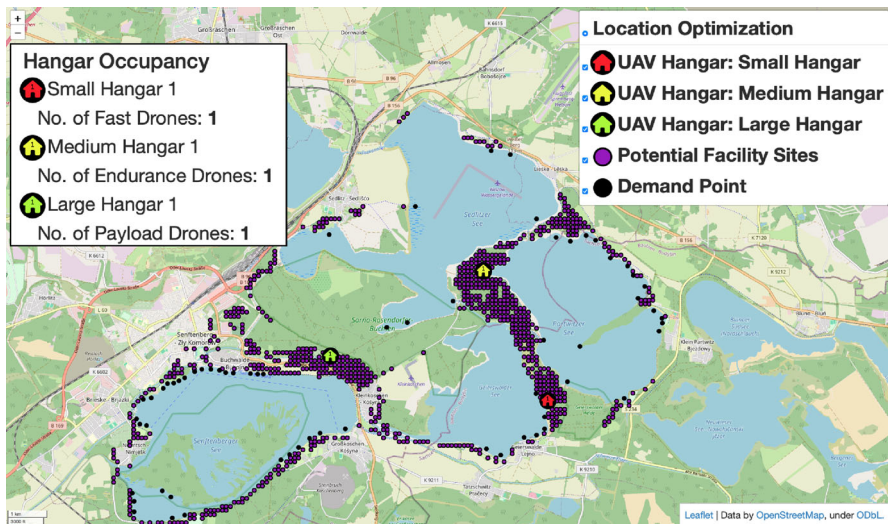


Fig. 10 CCMWMCLP model solution for the first use case with 1 to 4 hangars to be placed (from top left to bottom right)

each have different tasks: First, we consider a fast, light-weight UAV that is used to rapidly approach and localize the distressed swimmer. The second UAV is responsible for monitoring and surveillance and therefore features a long battery life. Another UAV is utilized to carry medical equipment, such as flotation devices, and must therefore have a high payload instead of being particularly fast or durable. So we define $\mathcal{C} = \{ \text{"localization"}, \text{"surveillance"}, \text{"medical delivery"} \}$ and contribution matrix $\mathcal{L}$ as identity matrix $\mathcal{L} = (l_{uc}) = I_3$. This means that each UAV is employed exclusively for one purpose. This can be adjusted by modification of $\mathcal{L}$. The remaining parameters are listed in Table 6. The instance was again generated using docplex and Python and was solved utilizing the same solver and computer settings from above but run into the time limit of 7200s with a relative optimality gap of 37.78%, and is visualized in Figure 11. This indicates that the model probably scales quite heavily with increasing instance size and requires further research in the future. Presumably, a reduction in the number of potential facility locations or demand points would already lead to a significant reduction in the computing time. In our case, the number of demand points $|\mathcal{D}| = 65$ is already sufficiently limited; however, the total number of potential facility

Table 6 Location optimization model parameters for the second use case with three UAV and hangar types, respectively

Name	Parameter	Value (CCMCLP)
Potential sites maximum capacity	$F_i^{\max}$	$6 \forall i \in \mathcal{F}$
UAV size	ψ_u	$\{1, 2, 3\}$
Available UAVs	$N_u^{\max}$	$\{1, 1, 1\}$
Hangar size	s_h	$\{1, 3, 5\}$
Available hangars	$P_h^{\max}$	$\{1, 1, 1\}$
Hangar delay	τ_h	$\{5s, 8s, 15s\}$
Operation altitude	v	$50.0m$
UAV battery capacity	σ_u	$\{720s, 1800s, 900s\}$
Battery threshold	T_{jc}	$\{300s, 900s, 300s\} \forall j \in \mathcal{F}, c \in \mathcal{C}$
Time threshold	$\hat{T}_{jc}$	$\{60s, 480s, 600s\} \forall j \in \mathcal{F}, c \in \mathcal{C}$
UAV maximum speed	$v_u^{\max}$	$\{33.333 \frac{m}{s}, 25.0 \frac{m}{s}, 11.111 \frac{m}{s}\}$
UAV acceleration rate	$a_u^{\max}$	$\{20.0 \frac{m}{s^2}, 5.0 \frac{m}{s^2}, 2.0 \frac{m}{s^2}\}$
UAV climb rate	c_u	$\{20.0 \frac{m}{s^2}, 5.0 \frac{m}{s^2}, 2.0 \frac{m}{s^2}\}$


Fig. 11 CCMCLP model solution for the second use case of three different UAV and hangar types, respectively

sites $|\mathcal{F}| = 834$ is disproportionately high relative to the number of locations to be selected $p = 1, \dots, 8$.

5.2 Search and Rescue Flight Trajectory Optimization

In the following, we will apply the flight planning MILP from Section 3 to the test area. Since water rescue is very time-critical, the design of the search path is particularly important. We describe the procedure in detail below.

5.2.1 Defining Waypoints for Search Area Coverage

The model operates with a discrete set of waypoints $\mathcal{W}$ to be scanned and determined in advance with respect to the search area and UAV- and operation-specific parameters. On the one hand, we are interested in minimizing the number of waypoints in order to keep the computational effort low (Fügenschuh et al. 2021). On the other hand, the UAV should cover the entire search area, otherwise it will miss the target. To address this issue, we introduce a Set Cover Problem (SCP) formulation as follows:

We define a discrete set of points $\mathcal{P}$ to be covered inside search area $\mathcal{A}$ and a set $\mathcal{S}$ of potential waypoints to cover the points from $\mathcal{P}$. Furthermore, we define a coverage matrix $\mathbf{C}^{\text{SCP}} = (c_{ip}) \in [0, 1]^{|\mathcal{S}| \times |\mathcal{P}|}$ with

$$c_{ip} = \begin{cases} 1, & \text{if point } p \text{ is covered from point } i, \\ 0, & \text{else,} \end{cases} \quad i \in \mathcal{S}, p \in \mathcal{P}. \quad (73)$$

To incorporate coverage, we introduce binary variables

$$x_i = \begin{cases} 1, & \text{if point } i \text{ is used,} \\ 0, & \text{else,} \end{cases} \quad i \in \mathcal{S}, \quad (74)$$

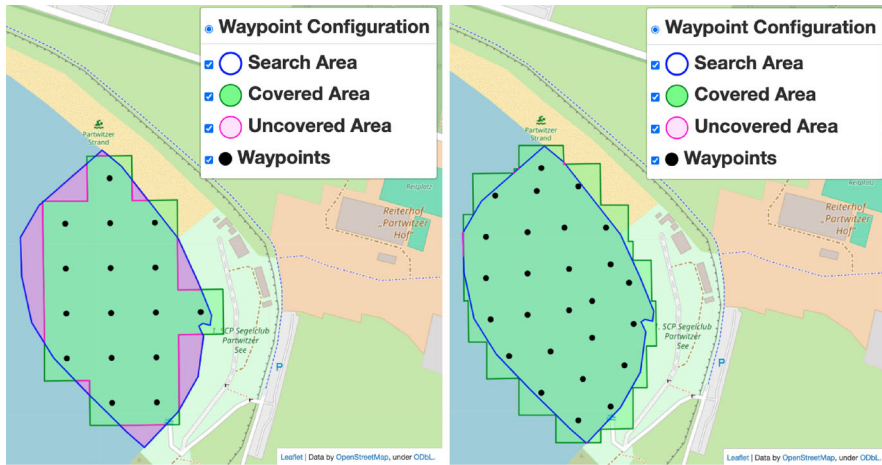
and constraints

$$\sum_{i \in \mathcal{S}} c_{ip} x_i \geq 1 \quad \forall p \in \mathcal{P}. \quad (75)$$

The objective now is to use as few points from $\mathcal{S}$ as possible:

$$\min \sum_{i \in \mathcal{S}} x_i. \quad (76)$$

In our application, we determine the sets $\mathcal{P} = \mathcal{S}$ by an equidistant grid approach with fixed grid spacing values $\Delta_x, \Delta_y \in \mathbb{R}_{>0}$ in coordinate directions x and y , respectively. The coverage highly depends on the operation altitude $v_{u,t}$ of UAV u at time step t . Here, we assume a fixed operation altitude $v_u = v_{u,t}$ for all $t \in \mathcal{T}_f$ and determine covered points depending on v_u and the UAV's angular aperture angles α_u and β_u or associated vertical and horizontal Field of View (FOV) $2a_u$ and $2b_u$. However, we cannot know in advance from which side a waypoint will be approached by the UAV. Consequently, it would be logical to consider a point covered, if it lies inside a circle with radius $\min\{\alpha_u, \beta_u\}$. However, experience has shown that this increases the number of waypoints immensely, which is why we have opted for covering with



(a) The equidistant waypoint approach with 16 waypoints and 83.74% coverage. (b) The set cover waypoint approach with 26 waypoints and 99.97% coverage.

Fig. 12 Equidistant (left) and set cover waypoint selection methods (right) at Lake Partwitz, utilizing $|\mathcal{P}| = |\mathcal{S}| = 1659$ points, solved with CPLEX 22.1.1.0 (International Business Machines Corporation 2023) on an Apple MacBook Pro with an Intel Core i9 running 1 thread at 2.4 GHz clock speed and 64 GB RAM within 1.23 seconds

squares with side length $\min\{2a_u, 2b_u\}$. This still guarantees very good coverage for the S&R operation.

We choose $\Delta_x = \Delta_y = 5.0$ meters, $\alpha_u = 45.0^\circ$, $\beta_u = 56.0^\circ$ and operation altitude $h_u = 60.0$ meters and applied the set cover approach to all search areas. The result is shown for an exemplary area in Figure 12, comparing the equidistant approach (see Figure 12a) to the set cover approach (see Figure 12b).

5.2.2 Model Solutions and Insights

We apply the flight planning MILP to some example use cases at the Lusatian Lake District. We consider a UAV S&R operation situated directly near the UAV hangar, first with a single UAV and secondly with an additional identical UAV. The instance parameters are listed in Table 7.

We now utilize two different score value methods: Firstly, we consider equal score values and secondly we consider a suspected accident site and utilize Gaussian distributed score values, the further away from the accident site, the lower the score value. This results in the four following S&R configurations: (1) Equidistant waypoint spacing with equal score values, (2) equidistant waypoint spacing with Gaussian score values, (3) set cover waypoint spacing with equal score values, (4) set cover waypoint spacing with Gaussian score values. The optimal solutions are shown in Figures 13 and 14, respectively.

In another experiment for the same search area, we added a restriction area in form of a beach. The optimal solutions are displayed in Figures 15 and 16. Note that in some images, such as Figures 13b or 14c, the flight trajectory seems to not be optimal in terms

Table 7 S&R operation parameter configuration for the flight planning model from Section 3

Name	Parameter	Value
Number of time steps	T	30
Number of fine time steps to get a coarse one	n_f	5
Time step length	Δ_t	5.0s
Maximum operational distance	$\delta_{u,w}$	1.0m
safety distance between UAVs	$\bar{\varepsilon}$	1.0m
Time window to visit waypoint	$\mathcal{T}_w$	$\mathcal{T} \forall w \in \mathcal{W}$
Maximum velocity	$v_u^{\max}$	65.0 $\frac{m}{s}$
Maximum acceleration	$a_u^{\max}$	4.0 $\frac{m}{s^2}$
Safety energy level	$g_u^{\min}$	10%
vertical aperture angle	α_u	45.0°
horizontal aperture angle	β_u	56.0°
battery consumption	$n_{u,j}$	0.6 $\frac{\%}{s}$
Objective scaling value	M^{obj}	100



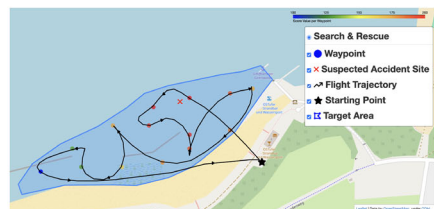
(a) Equidistant waypoint spacing with equal score values.



(b) Equidistant waypoint spacing with Gaussian score values.



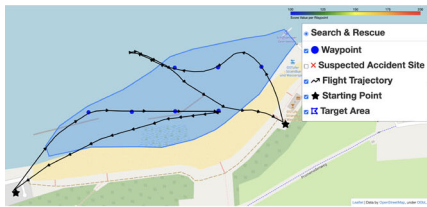
(c) Set cover waypoint spacing with equal score values.



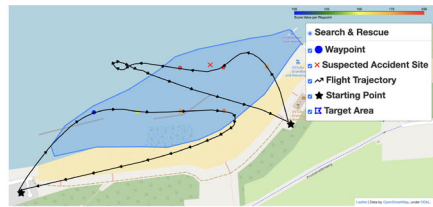
(d) Set cover waypoint spacing with Gaussian score values.

Fig. 13 Single UAV S&R operation path planning for an example area at the Lusatian Lake District

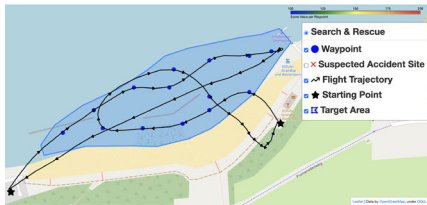
of time. This is a relic of the two-level time discretization approach, since the variable $\theta^{\max}$, which measures the maximum flight time, can only be determined for coarse time steps $\mathcal{T}$. Therefore, if an improvement up to the previous coarse time step is not possible, there is no more improvement possible in the objective function. In addition, the focus of the objective function lies very much on the rapid collection of waypoints. In Figures 14a and 14b can be seen that the UAV collects the waypoints very quickly and then needs a little more distance to decelerate from the high speed and fly in the



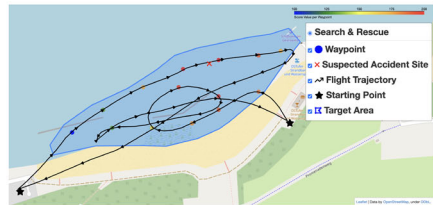
(a) Equidistant waypoint spacing with equal score values.



(b) Equidistant waypoint spacing with Gaussian score values.

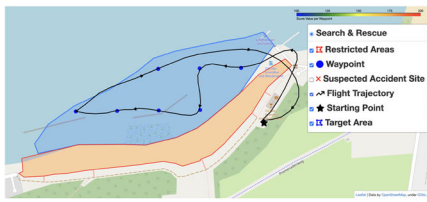


(c) Set cover waypoint spacing with equal score values.

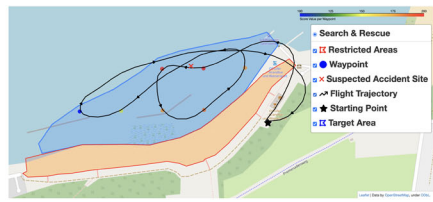


(d) Set cover waypoint spacing with Gaussian score values.

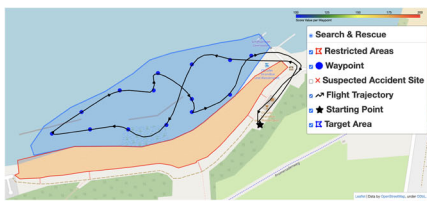
Fig. 14 S&R operation path planning with two identical UAVs for an example area at the Lusatian Lake District



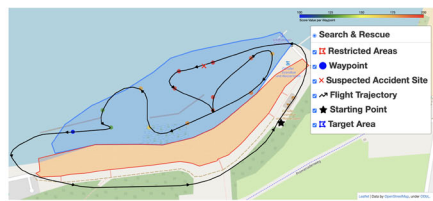
(a) Equidistant waypoint spacing with equal score values.



(b) Equidistant waypoint spacing with Gaussian score values.



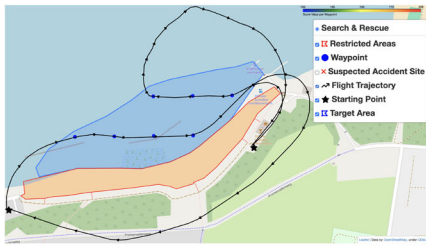
(c) Set cover waypoint spacing with equal score values.



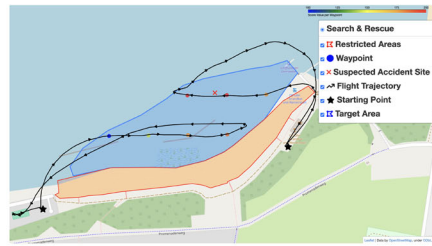
(d) Set cover waypoint spacing with Gaussian score values.

Fig. 15 Single UAV S&R operation path planning for an example area at the Lusatian Lake District including a beach as restricted area

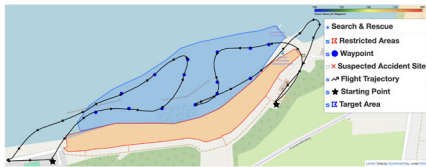
opposite direction back to the end point. It would also be conceivable to control the speed at which waypoints are collected by introducing additional constraints. Note that in contrast to the Traveling Salesman Problem (TSP), overlaps in the route trajectory can occur here due to the objective function and the flight dynamics of the UAVs. Nevertheless, collision avoidance is guaranteed at all times.



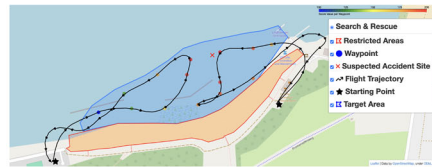
(a) Equidistant waypoint spacing with equal score values.



(b) Equidistant waypoint spacing with Gaussian score values.



(c) Set cover waypoint spacing with equal score values.



(d) Set cover waypoint spacing with Gaussian score values.

Fig. 16 S&R operation path planning with two identical UAVs for an example area at the Lusatian Lake District including a beach as restricted area

6 Conclusions

In this paper, we presented a comprehensive framework for UAV-assisted water rescue, introducing an Unmanned Aircraft System (UAS) consisting of drones or Unmanned Aerial Vehicles (UAVs) that are stored in UAV hangars. Our scope of work included mathematical formulations of a UAV flight pre-planning MILP model and a location-allocation optimization MILP for the proposed UAS and application to a real-world inland water area in Germany.

The flight planning MILP model proved to be a powerful tool for detailed advance planning of S&R operations of heterogeneous swarms of UAVs. Our results showed that very precise flight planning with smooth and realistic flight trajectories is possible, taking into account Newtonian flight dynamics, restriction areas, prioritization of waypoints, battery restrictions and wind. Waypoint locations are decisive for coverage of the search area, hence we introduce a set cover approach to improve coverage compared to an equidistant grid approach. However, this increases the number of waypoints and therefore also the computational effort to solve the model instance. These undesirable scaling effects need to be investigated in more detail in future research. Another possible research topic is to investigate the performance of the flight planning MILP has advantages over established techniques, taking into account real-time planning and UAV-swarm intelligence. An aspect that has not yet been considered is the varying flight altitudes of UAVs during search operations, which result in different fields of view. Additionally, the incorporation of rotatable cameras or cameras with optical zoom could be explored. Another challenge in our flight planning model is to eliminate the corner-cutting phenomenon when flying around restricted areas. For very large instances, e.g. covering very large search areas including restricted areas

with a large swarm of UAVs, it may well be necessary to develop additional heuristic methods. Further, implementation of a live flight planning mechanism is essential for practical applications and was not addressed in this paper.

The introduced location-allocation optimization MILP model is a valuable tool for well-founded positioning and sizing of available and to-be-acquired limited resources (UAVs and hangars) in order to cover the area-specific demand, shorten UAV approach times and ultimately improve the chances of a successful rescue mission. The model enables evaluation of the suitability of different UAVs and hangar types and integration of different purposes. Coverage is measured on the one hand in terms of how quickly the potential swimming areas are reached and on the other hand in terms of the remaining battery capacity to conduct the S&R mission or other activities on site.

Our results have shown that the model finds reasonable UAS configurations that are customized to the individual demand. However, some of the experiments indicate poor scaling effects for large-scale instances, which in turn leads to limitations in practical application. This is caused by the rapidly growing number of optimization variables and constraints in the model, which can be influenced in particular by the number of demand points and potential facility sites, as well as by the number of UAV types and hangar types. It is therefore advisable, if possible, to perform some preliminary screening in order to reduce the instance size. Some effective measures include reducing the set of potential facility sites to an appropriate size relative to the number p of sites to select, or cluster individual nearby demand points and use the cluster centroids in the model. If the UAVs serve only a few demand points jointly, for example when deploying very few UAVs or hangars in a very large operational area, dividing the area into several individual instances could also be considered. The exact extent of the scaling issues and how to counteract this will be the subject of further research in the form of extensive computational studies.

Further, exploring different types of signal functions is useful to customize the appropriate incorporation of UAVs cooperation for specific use cases. Also, reducing computational time is a desirable goal. Furthermore, in use cases with a lot of no-fly zones, measuring distances by the euclidean distance might be insufficient, so the path planning MILP can be adapted by simply leaving out all waypoints.

In order to investigate the influence of different UAS configurations and parameters on the S&R mission outcome, a simulation approach to examine the combined location-allocation optimization and flight route planning is useful. This would allow to simulate employment of different UAS configurations and compare their performances. In addition, the extent to which drowning swimmers can be reliably detected using image processing methods is to be investigated. The extension of the framework to include simulation and drowning swimmers detection components will be subject of the second part of this publication.

The developed framework is versatile and its applicability to different use cases needs to be investigated. The most promising use cases are the following: The UAS can be utilized to serve as first response to reported wildfires and can identify fire origins or embers and transmit situation reports to the fire department. It can also improve S&R operations in remote or challenging terrains, such as providing medical care and rescuing injured hikers and climbers in mountainous tourism regions. The transportation of AEDs can also be considered. In addition, the location-allocation

model may find application to EMS systems in general, this time with various vehicles as resources, such as helicopters, UAVs, ambulances and medical transport vehicles allocated to rescue stations of various sizes and functionality.

Overall, our results suggest that the developed framework is promising for enhancing swimmer's safety in unguarded swimming areas, highlighting several key benefits of using UAVs in water rescue operations. They offer unprecedented opportunities to quickly cover large areas, reach inaccessible locations and provide real-time sensory data to emergency responders. Our system has the potential to have a significant impact on emergency response to water-related incidents, enhancing swimmer safety in open water swimming areas aiding professional lifeguards with their challenging task of saving human lives.

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Data Availability The authors declare, that upon reasonable request, the data and the code are available from the corresponding author.

Declarations

Ethics approval This study did not involve human participants or animals. Therefore, ethics approval and consent to participate are not applicable.

Consent for publication This manuscript does not contain any individual person's data in any form (including images, videos, or identifiable details); therefore, consent for publication is not applicable.

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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